

IMST sentire™ Radar Module DK-sR-1200 Development Kit User Manual

Version 1.11



About this Document

This document is a manual of the sR-1200 Development Kit including IMST's sentire™ Radar Module. The following table displays the changes applied to this manual and assigns a version number to each document release.

Version	Date of Change	Reason of Change
1.0	2016-03-16	initial document
1.1	2016-05-02	modified specification
1.2	2016-05-31	modified GUI layout
1.3	2016-06-23	added description of the Range Finder algorithm
1.4	2016-07-25	data saving and replay description modified
1.5	2016-11-15	modified ethernet pin description
1.6	2016-11-22	modified SenTool ethernet settings
1.7	2017-03-17	modified specification
1.8	2017-06-07	modified SenTool description to support version 1.3
1.9	2017-08-03	modified SenTool description to support version 1.3.1
1.10	2017-11-01	added section 2.5.3 HTTP Server
1.11	2018-03-22	modified section 2.5.3 HTTP Server

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1. Introduction

1.1 Product Description

This manual describes the IMST's sentire™ Radar Module and its installation procedure. The radar operates in FMCW or CW mode at a frequency of 24 GHz and provides the capability for range and velocity measurements in a wide area of applications. Using the integrated transmitter and receiver antennas, it is also possible to estimate the angle-of-arrival of the back-scattered signal, corresponding to a certain target direction. The radar module is available with three different interfaces (SPI, CAN or Ethernet). In case of the SPI interface, an external SPI-to-USB adapter allows to connect the radar directly to the computer. While an external 12 V power supply is required for powering the SPI or CAN bus modules, the Ethernet version uses PoE supply. The unit with Ethernet interface has CE type approval for operation in the ISM band ranging between 24 GHz to 24.25 GHz at an output power of 19 dBm EIRP (for more details please adhere to the national frequency regulations of your country). The technical specifications of the sentire™ Radar Module are summarized in chapter 4. Independently from the selected interface, the module provides an extensive set of user commands for the following tasks:

- configuration of radar parameters, as the operation mode or the frequency range,
- accessing internal radar parameters, as the current frontend temperature or the transmission power,
- triggering a measurement process and accessing the generated data,
- starting high-level applications, e.g. the Human Tracker algorithm.

However, this document doesn't contain an exhaustive explanation of all commands supported by the Radar Module. In order to access all functions of the sensor, the user can refer to the command description listed in the command interface manual /CMD/.

Additionally, the program IMST SenTool is provided as part of the Development Kit (see chapter 3). This software offers the possibility to modify the most of the radar parameters, to make different measurements as well as to visualize and save measurement data. The IMST SenTool is very helpful to explore the possibilities of the sentire™ Radar Module or to determine suitable settings for a particular application or operation.

1.2 Important Information

The following section contains useful hints and recommendations for a save operation of the sentireTM Radar Module.

1.2.1 Electrical Safety

The sentireTM Radar Module operates at a maximum DC voltage of 54 V (PoE version, refer to chapter 4), which falls under the IEC definition of Extra-Low Voltage (ELV). Due to the non-conducting housing, no additional protection against electrical shock is required. In order to avoid damaging the sensor, it is equipped with an fuse of 0.5 A and a reverse voltage protection. Additionally, the following rules have to be followed:

- operate the device only with specified DC voltage levels,
- don't open the housing while the device is powered,
- don't intentionally change the polarity.

1.2.2 ESD Protection

The sentireTM Radar Module is protected against Electrostatic Discharge (ESD) and tested according to DIN EN 61000-4-2. While working with an open housing (e.g. connecting a new cable to the Interface board), the following rules have to be followed:

- the device must be unpowered,
- the cable must be disconnected,
- an electrical grounding is required.

1.2.3 RF Safety Considerations

The sentireTM Radar Module operates in the ISM band of 24 GHz and generates a maximum RF output power of 10 mW (10 dBm), corresponding to an EIRP power of approximately 19 dBm. With an active radome surface of 8 cm × 8 cm, the maximum RF power density is expected to be 1.56 W/m², which is significantly lower than the threshold value of 10 W/m², defined by the International Commission on Non-Ionizing Radiation Protection (ICNIRP) in 1998.

However, if several RF devices operate simultaneously in the same frequency range, different crosstalk effects can appear, leading to a reduced sensor sensitivity or erroneous measurement data. It is at the responsibility of the user to take care of these effects.

1.2.4 Thermal Operating Conditions

The housing of the sentire™ Radar Module is made of black Polycarbonate, is classified as flame-retardant and approved to the following standards:

- V-2 material class of the UL94 and IEC 60696-11-10 standards,
- B2 fire class of the DIN 4102 standard.

Additionally, the following limitations must be taken into account:

- operate the radar only in the temperature range specified in section 4,
- long term measurements at high temperatures stress the RF components and may affect the sensor's lifetime,
- internal temperatures of more than 85° temporary will deactivate the radar frontend,
- please provide additional cooling in high-temperature environments.

1.2.5 Mechanical Safety

The housing of the sentire™ Radar Module consists of two solid parts: a box containing the radome and PCB mounting structures, and a lid combined with a mounting plate. Both parts are connected with four screws. Due to a rubber sealing ring between box and lid, the radar is protected against splash water and is compliant to the IP65 standard. In order to ensure a save operation of the sensor, the following rules have to be followed:

- always use all mounting holes of the housing to fix the radar,
- don't use the attached cable for mounting reasons, don't pull the cable,
- don't mechanically stress the radar (e.g. pressure, shock, strong vibrations),
- don't cover the RF active side of the sensor (radome) with tape or other materials,
- don't use the radar if the housing is damaged.

1.2.6 Chemical Conditions

The material used for the housing of the sentire™ Radar Module is chemically resistant against dilute acids, alcohols and many greases and oils. It is stabilized against exposure to UV radiation and provides an excellent weathering resistance.

2. Installation

2.1 Hardware Overview

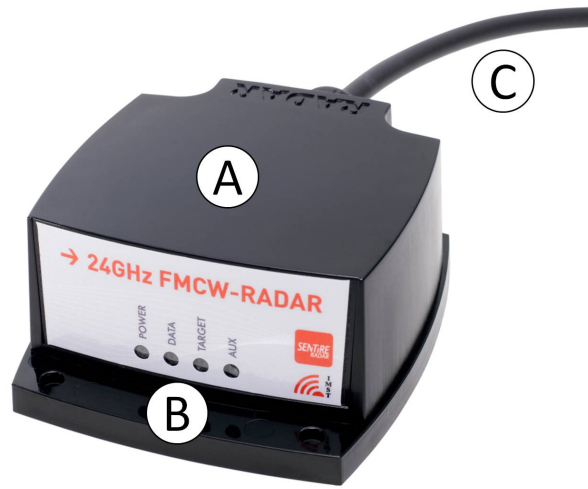


Figure 2.1: IMST sentireTM Radar Module (A: radome and antenna position, B: LED indicators, C: interface and power supply cable)

The sentireTM Radar Module (see figure 2.1) is a highly integrated device and consists of three hardware groups mounted into a robust and compact plastic housing:

- *Interface Board*, containing the power supply and the communication module (SPI, CAN or Ethernet),
- *Baseband Board*, containing the built-in digital signal controller (DSC) and a data acquisition unit,
- *Frontend Board*, containing the RF components as well as the integrated patch antennas.

Interface, Baseband and Frontend boards are stuck together using multi-polar connectors. The housing is almost completely transparent in the radiated frequency band and also acts as radome to protect the sensitive antenna structures.

To visualize the current operation step, the radar provides four LEDs:

- *POWER* indicates that the module is connected to power supply and is ready for operation,
- *DATA* indicates that a communication process between radar and host is in progress (e.g. transmission of user commands or measurement data of any type),
- *TARGET* indicates that a target object has been detected by a high-level algorithm,
- *AUX* indicates unexpected radar behavior, e.g. in case of an internal error.

On the back side, the sentireTM Radar Module provides a multi-functional cable for power supply and communication interface (see figure 2.1 C). Depending on the built-in Interface board, the radar supports one of three bus systems: SPI, CAN or Ethernet (for more details please refer to sections 2.3 to 2.5).

2.2 Computer Requirements

Table 2.1 lists the computer requirements for operating the sentireTM Radar Module using the SenTool software. For modules with SPI interface, the corresponding SPI-to-USB adapter is integrated in the data cable. For modules with CAN interface, an additional CAN-to-USB converter is required (sysWorxx USB-CANmodul1, not included).

Parameter	Minimum Requirement
Operation System	Windows 7 or higher
Free HDD Space	100 MB
CPU	Dual Core 2.5 GHz (recommended)
Free RAM	150 MB (1 GB recommended)
USB (SPI or CAN module)	USB2.0 or higher
Network (Ethernet module)	LAN with 10 Mbit/s or higher, PoE

Table 2.1: Hardware requirements for SenTool installation

2.3 SPI Interface

A sentireTM Radar Module equipped with an SPI interface can be generally accessed via two different methods:

1. direct SPI access using the connection pins described in section 2.3.1,
2. using the provided SPI-to-USB converter (described in section 2.3.3) and the corresponding FTDI computer library.

2.3.1 SPI Pin Description

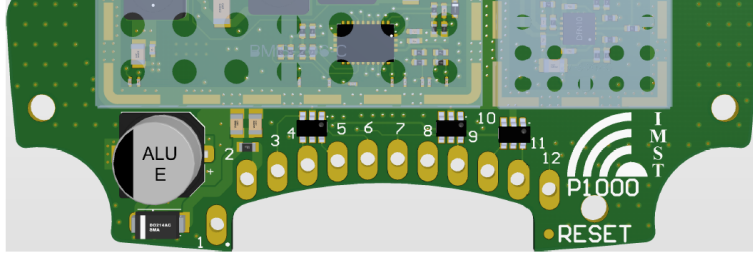


Figure 2.2: Connection pins of the SPI interface board

Pin	Name	min. [V]	max. [V]	Description
1	GND	—	—	Ground for power supply
2	VCC	10.5	13.0	Power supply voltage
8	SPI_LVL	1.65	5.5	SPI Logic Level (INPUT)
9	SPI_CLK	-0.5	5.5	SPI Clock Frequency (INPUT)
10	SPI_MOSI	-0.5	5.5	SPI Master Data Out (INPUT)
11	SPI_MISO	-0.5	5.5	SPI Master Data In (OUTPUT)
12	SPI_CS	-0.5	5.5	SPI Chip Select (INPUT)

Table 2.2: Pin description of the SPI interface board

Figure 2.2 shows the connection pins of the SPI interface board. The pins which are relevant for DC power supply and for the SPI interface are listed and described in table 2.2. All other pins (3 to 7) serve for development and debugging reasons and should not be connected to any external voltage.

2.3.2 SPI Parameters

Parameter	Value
SPI Mode	Master
SPI Clock Rate	0.1 – 2 MHz
SPI Chip Select	low active
SPI Master Transmit Data	on falling edge
SPI Master Receive Data	on rising edge
SPI Word Length (read & write)	16 bits
SPI Byte Order (read & write)	LSB first

Table 2.3: SPI configuration parameters for the controlling system

The SPI interface of the sentire™ Radar Module is directly connected to the DSC and operates in slave mode (i.e. the controlling system is SPI master). The outside TTL

level of the SPI interface pins can be configured using the pin SPI_LVL (see table 2.2) in order to enable a two-directional translation between any of the 1.8 V, 2.5 V, 3.3 V or 5.5 V voltage nodes. Table 2.3 shows a list of parameters required for configuration of the controlling system.

2.3.3 SPI-to-USB Converter

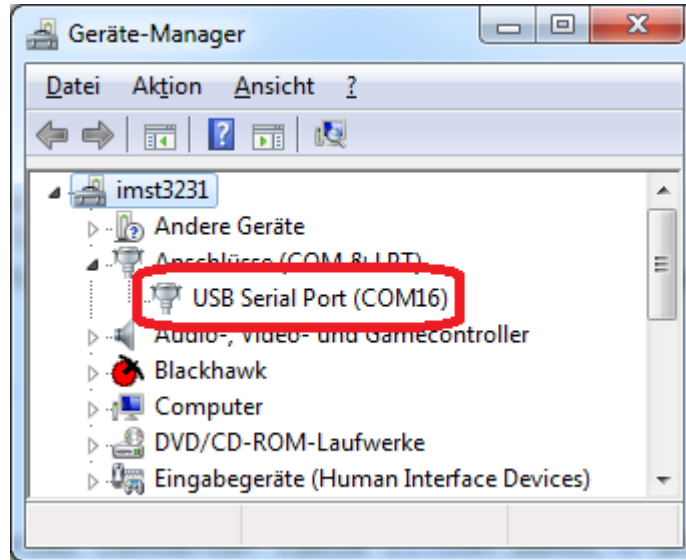


Figure 2.3: Virtual COM port appearing in the Windows Device Manager after connecting the TTL-to-USB Serial Converter Cable

The SPI version of the sentireTM Radar Module is optionally delivered with the TTL-to-USB Serial Converter Cable C232HM-DDHSL-0 which is based on the FTDI chip FT232H (for more details please refer to www.ftdichip.com). In the present application, the FTDI chip is configured to act as SPI-to-USB converter between the Radar Module and the controlling computer. If connected to the USB port of a computer, the FTDI chip creates a virtual COM port (see figure 2.3) which can be accessed using FTDI's driver interface FTD2XX.dll. Please note, that the FTDI driver must be installed in advance before using the USB port.

Additionally, IMST provides a Python user interface package which is able to communicate with the Radar Module through the FTDI library (see /CMD/). The software IMST SenTool (available as part of the Development Kit) supports this communication interface. In order to allow the user a quick start with programming the Radar Module, a simple Python example code is also provided.

2.4 CAN Bus Interface

Similarly to the SPI version, a sentireTM Radar Module equipped with a CAN interface can be also be accessed in two general ways:

1. direct CAN access using the connection pins described in section 2.4.1,
2. using an optional CAN-to-USB converter (described in section 2.4.3).

2.4.1 CAN Pin Description

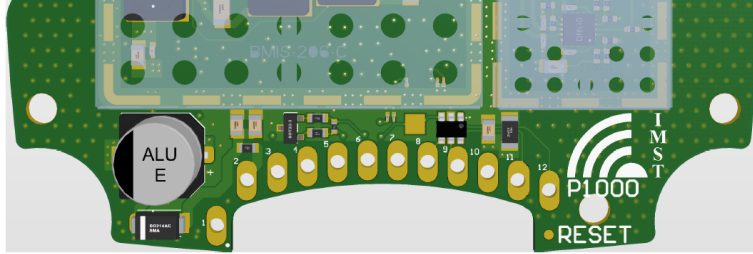


Figure 2.4: Connection pins of the CAN interface board

Pin	Name	min. [V]	max. [V]	Description
1	GND	—	—	Ground for power supply
2	VCC	10.5	13.0	Power supply voltage
5	CANH	-2.0	7.0	High level CAN bus line
6	CANL	-2.0	7.0	Low level CAN bus line
3, 4, 7, 10	GND_S	—	—	Shielding ground

Table 2.4: Pin description of the CAN interface board

Figure 2.4 shows the connection pins of a CAN bus version of the Radar Module, table 2.4 provides a description of the corresponding functionality. The DC power supply pins 1 and 2 are similar to the SPI version, while the pins 5 and 6 provide a two-wire CAN bus interface corresponding to ISO 11898-2 (Highspeed-CAN standard). The use of connection pins indicated as GND_S is optional (for long range transmission, these pins can be connected to the cable shielding). All other pins serve for development and debugging reasons and should not be connected to any external voltage.

Internally, the CAN bus data lines are terminated using a $120\ \Omega$ resistor. Depending on the final CAN network configuration, it may be necessary to remove the resistors from some modules in order to ensure an appropriate network impedance.

2.4.2 CAN Parameters

When connected via CAN bus interface the sentireTM Radar Module can be addressed using its node ID (which should be unique in the CAN network) as well as a broadcast ID which is identical for all radar modules. After a successful connection, the CAN bus parameters of a radar module can be accessed and optionally modified as described in the Communication Interface manual (see /CMD/). Table 2.5 summarizes the CAN bus configuration parameters.

Parameter	Value
Baud Rate [kbaud]	10, 20, 50, 100, 125, 250, 500, 800, 1000
Default Baud Rate [kbaud]	1000
CAN Node ID	0 to 62 (default: 2)
Broadcast CAN Node ID	63
CAN Word Length (read & write)	16 bits
CAN Byte Order (read & write)	LSB first

Table 2.5: CAN configuration parameters of the Radar Module

2.4.3 CAN-to-USB Converter

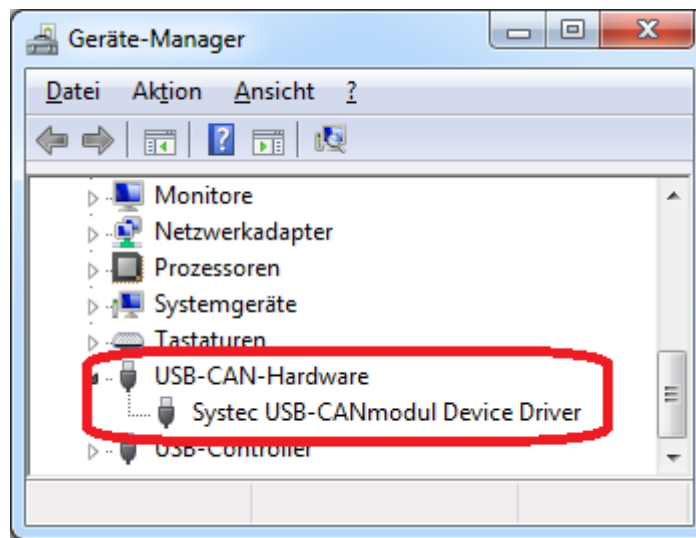


Figure 2.5: USB-CAN device appearing in the Windows Device Manager after connecting the Systec CAN-to-USB Converter

In order to connect a sentireTM Radar Module to a computer, an optional CAN-to-USB converter can be applied. IMST recommends to use Systec's compact USB-CANmodul1 which acts as translator between the USB port and the CAN network containing one or several radar sensors (for more details please refer to www.systec-electronic.com). If connected to the USB port of a computer, the USB-CANmodul1 appears in the Windows Device Manager as shown in figure 2.5. Please note, that the corresponding driver must be installed in advance before using the USB port.

Additionally, IMST provides a Python user interface package which is able to communicate with the Radar Module through the presented CAN-to-USB converter (see /CMD/). The software IMST SenTool (available as part of the Development Kit) supports this communication interface. In order to allow the user a quick start with programming a CAN bus Radar Module, a simple Python example code is also provided.

2.5 Ethernet Interface

The sentire™ Radar Module can be connected to a LAN using its Ethernet interface. Even if the Radar Module in the most cases will be delivered with an attached cable, section 2.5.1 contains a pin description of the Ethernet Interface board. The corresponding communication parameters are listed in section 2.5.2.

2.5.1 Ethernet Pin Description

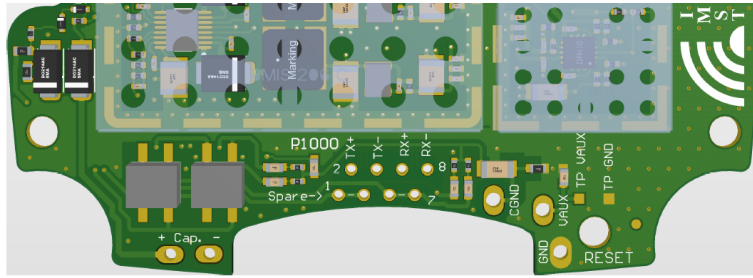


Figure 2.6: Connection pins of the Ethernet interface board

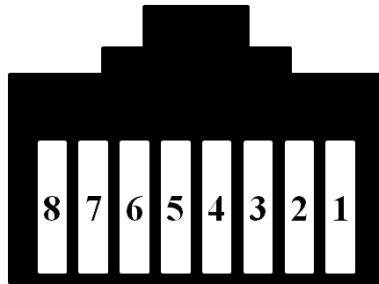


Figure 2.7: Pin numbering of a RJ-45 Jack (frontview)

The Ethernet version of the Radar Module is powered over the data lines or over the unused data lines of the 8-wire Ethernet cable and is compliant to the IEEE 802.3af PoE standard. Table 2.6 lists the relation between the pins of a RJ-45 Jack with a T568B pinout and the pins of the interface board. Since the IEEE 802.3af standard defines several possibilities to connect the DC power supply voltage to the Ethernet data lines, different modes are mentioned in Table 2.6. One of the following defined modes for the pins of the interface board can be chosen, where mode B1 is the standard mode:

- pins 2+4 $\rightarrow$ GND, pins 6+8 $\rightarrow$ 48 V (PoE Mode A1),
- pins 6+8 $\rightarrow$ GND, pins 2+4 $\rightarrow$ 48 V (PoE Mode A2),
- pins 1+3 $\rightarrow$ GND, pins 5+7 $\rightarrow$ 48 V (PoE Mode B1),
- pins 5+7 $\rightarrow$ GND, pins 1+3 $\rightarrow$ 48 V (PoE Mode B2).

Additionally to the interface data lines, the cable shielding should be connected to the pin CGND on the interface board.

Pin (Jack)	Pin (Interface Board)	Name	Colour Coding	Description
1	2	Tx+	 orange striped	Tx+ data line, 48 V/GND (PoE Modes A2/A1)
2	4	Tx-	 orange	Tx- data line, 48 V/GND (PoE Modes A2/A1)
3	6	Rx+	 green striped	Rx+ data line, GND/48 V (PoE Modes A2/A1)
4	3	Spare	 blue	48 V/GND (PoE Modes B2/B1)
5	1	Spare	 blue striped	48 V/GND (PoE Modes B2/B1)
6	8	Rx-	 green	Rx- data line, GND/48 V (PoE Modes A2/A1)
7	5	Spare	 brown striped	GND/48 V (PoE Modes B2/B1)
8	7	Spare	 brown	GND/48 V (PoE Modes B2/B1)

Table 2.6: Pin relation between RJ-45 jack and the interface board with T568B convention (graphics from Wikipedia)

2.5.2 Ethernet Parameters

Parameter	Value
Default Port	1024
Default IP Address	192.168.0.2
Gateway	192.168.0.1
Subnet Mask	255.255.0.0

Table 2.7: Ethernet configuration parameters of the Radar Module

When connected via Ethernet interface, the radar can be addressed using its Ethernet configuration parameters (see table 2.7). After a successful connection, the communication parameters of a radar module can be accessed and optionally modified as described in the Communication Interface manual (see /CMD/). One host can be connected to several radar-modules, when they have different connection parameters (port and IP address). The default connection parameters are valid 10 seconds after power-up. After this time the Radar Module is reconnected with the user-defined connection parameters.

Additionally, IMST provides a Python user interface package which is able to communicate with the Radar Module through the Ethernet interface (see /CMD/). The software IMST SenTool (available as part of the Development Kit) supports this communication interface. In order to allow the user a quick start with programming an Ethernet Radar Module, a

simple Python example code is also provided.

2.5.3 HTTP Server

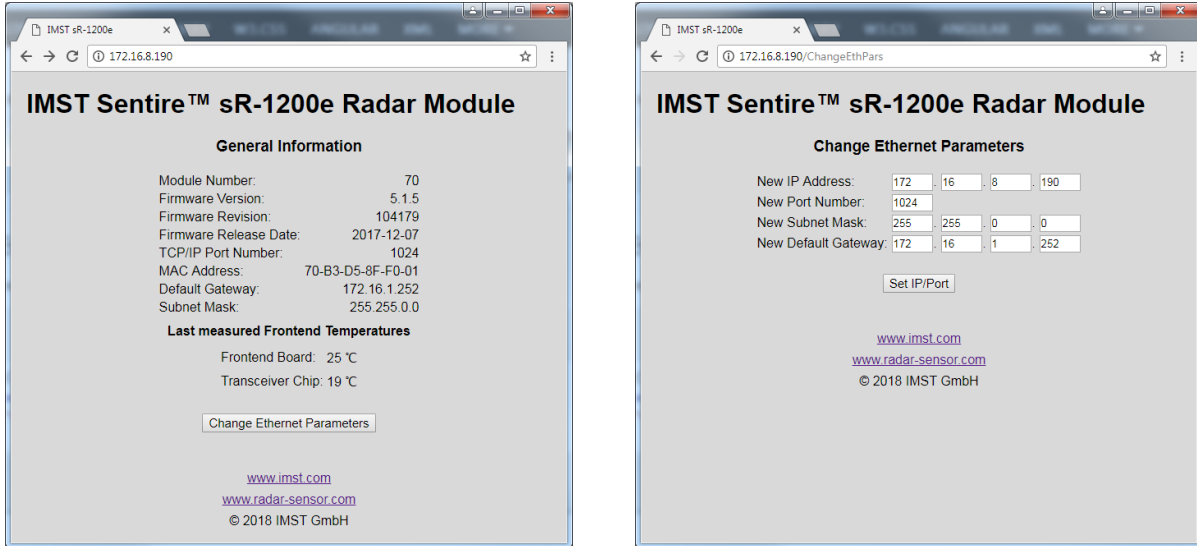


Figure 2.8: HTTP information page (a) and page to change IP address and Port (b)

The sentireTM Radar Module with Ethernet interface also runs a small HTTP server on the default port number 80. This means the radar will answer on a request from e.g. a browser to its standard IP address. The answer is a *HTML* page with module specific information like in figure 2.8 a. When clicked on the button *Change Ethernet Parameters* a page with the possibility to change the radars IP address and Port will be shown (see figure 2.8 b). The new IP address and Port can then be set by clicking the button *Set IP/Port*. Dependent on the used browser a message will pop up which informs if the setting was successful or not. There are some additional possible request URLs for specific operation which are described in /CMD/.

Note: If the IP address of the module is unknown, it is possible to get and change the current IP and Port configuration via a browser. Therefore the host computer should be in the same subnet (192.168.0.3 - 192.168.0.255) as the default IP address of the radar. Within the first 10 seconds after a power reset of the radar, the URL "192.168.0.2/ChangeEth-Pars" should be called by a browser. This page will show the current saved configuration. It is also possible to change these values here within the 10 seconds limit.

2.6 DC Power Supply

The sentireTM Radar Module equipped with SPI or CAN interface is powered using an external 12V DC power adapter (provided with the sensor). The power adapter has an 2-pin connector which is *not* protected against wrong polarity. As the radar is equipped with a reverse voltage protection, wrong DC polarity will not damage the sensor, but an

operation will be not possible in this case (a correct power supply is indicated by the *POWER* LED as described in section 2.1). Alternatively, the user can apply his own DC power supply to the radar sensor, as long as it complies to the specification listed in chapter 4.

3. IMST SenTool

The following chapter describes the graphical user interface IMST SenTool, which is enclosed to the Development Kit of the sentireTM Radar Module. This software visualizes the data received from the Radar Module by using its essential functions. In addition to basic radar configuration controls, different data plots and functions are implemented in the GUI to demonstrate the user all possibilities provided by the sensor. It also shows examples for post-processing of measurement data, like the *History Plot* (see section 3.7). In order to access the whole functionality of the Radar Module beyond the limitations of the IMST SenTool, the Command Interface Manual (see /CMD/) can be referred to communicate with the sensor in any programming language.

Figure 3.1 shows an example screenshot of the GUI with four highlighted functional groups, which are briefly discussed below. A more detailed discussion of these control elements is covered by the following sections.

1. *Controls*: Here the most important controls for the radar module and the GUI

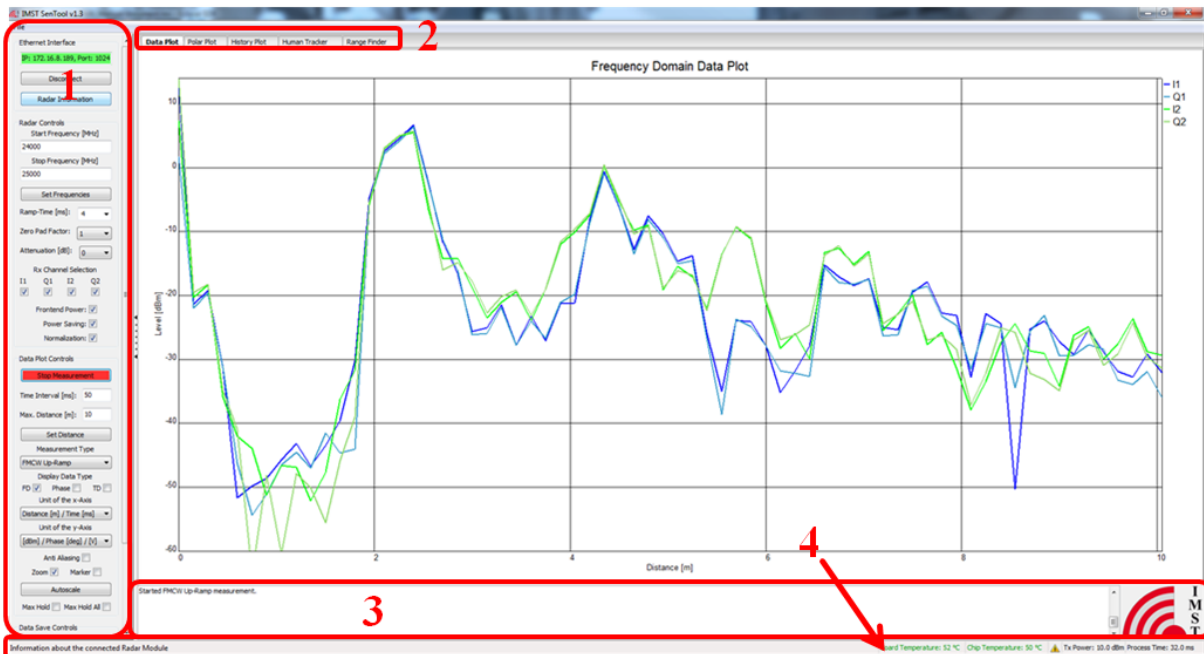


Figure 3.1: Screenshot of the IMST SenTool GUI (1: control panel, 2: data visualization tabs, 3: message window, 4: status bar)

are located. They are again separated into different fields which will be deeper described in the next sections. The field *Data Plot Controls* depends on the chosen tab (functional field 2 of figure 3.1) and changes by switching to another tab.

2. *Tabs*: Each of these tabs shows a different type of data plot or other radar functionality. A more detailed description is provided by sections 3.5 to 3.9. When another tab is selected by the user during a running process, the actual radar measurement is stopped.
3. *Message Window*: In this window all user activities and errors are displayed. The message history can be exported to generate a log-file.
4. *Status Bar*: The status bar is divided into different subsections and contains some information about the Radar Module, the GUI and the currently running measurement.

The current state of all GUI control elements is saved to a configuration file in the folder where the software has been installed (please note that this file cannot be edited by the user). This way, the settings of all data plots as well as all connection parameters will be restored at the next start of the program. In contrary, the internal parameters of the Radar Module are not saved to the configuration file, as the sensor itself writes a list of the most relevant parameters to EEPROM.

3.1 Connecting a Radar Module

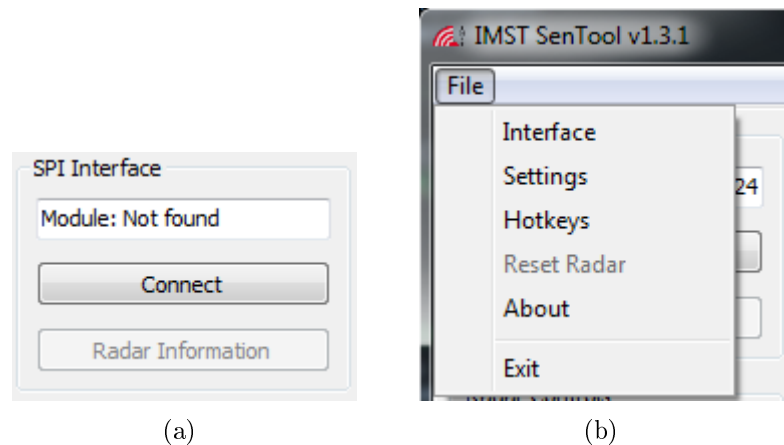


Figure 3.2: (a) Interface field in Controls, (b) Interface entry in the File menu

As already mentioned in section 2.1, the communication between host computer and sentireTM Radar Module can be established using three different interfaces: SPI, CAN or Ethernet. The IMST SenTool supports all of these interfaces. At the first start of the software or when another radar module is used, the connection type as well as the communication parameters must be entered by the user. Figure 3.2a shows the interface control field at the first start of the GUI. Figure 3.2b shows how the interface dialog can

be entered via the File menu. Alternatively, it can be called by a double-click at the text field of the interface control. Please note that the interface menu can only be entered while a radar module is disconnected from the GUI. Similarly, all other functions are blocked while no module is connected. It should also be mentioned that the power supply of the frontend is automatically switched ON after a radar module is successfully connected and switched OFF when a module is disconnected.

3.1.1 SPI Settings

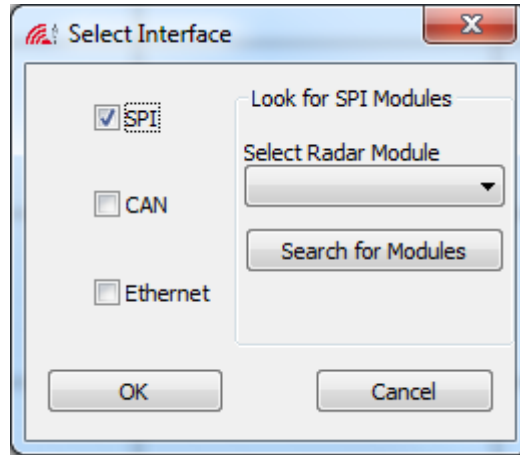


Figure 3.3: Settings of the SPI interface

The settings of the SPI interface can be configured by selecting the checkbox *SPI* of the *Select Interface* dialog (see figure 3.3). After pressing the button *Search for Modules*, the software starts searching for available radar modules connected to the computer via the SPI-to-USB adapter (see section 2.3). For each detected device, a corresponding identifier string appears in the droplist *Select Radar Module*. After one radar module has been selected from list, the configuration can be accepted by *OK* or rejected by *Cancel*. Finally, the selected Radar Module can be connected via the *Connect* button of the interface control field.

3.1.2 CAN Bus Settings

The settings of the CAN interface can be configured by selecting the checkbox *CAN* of the *Select Interface* dialog (see figure 3.4). In order to connect a Radar Module via a CAN-to-USB adapter (see section 2.4), its Node ID and Baud Rate must be entered into the corresponding control fields. If these parameters are unknown, the *Auto Detect* button can be pressed to search for available sensors (please note that this can take some time, as many parameter combinations are tried). Finally, the selected configuration can be accepted by *OK* or rejected by *Cancel*.

Additionally, the connection parameters of a radar module can be modified by pressing the button *Set new Parameters*. Hereby, the currently entered Node ID and Baud Rate will be adopted and saved into the sensor's EEPROM.

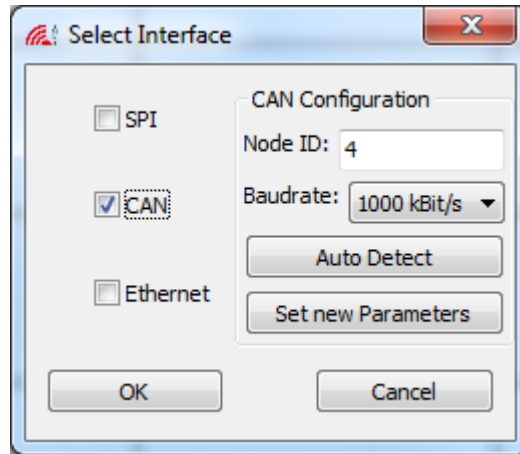


Figure 3.4: Settings of the CAN Bus interface

3.1.3 Ethernet Settings

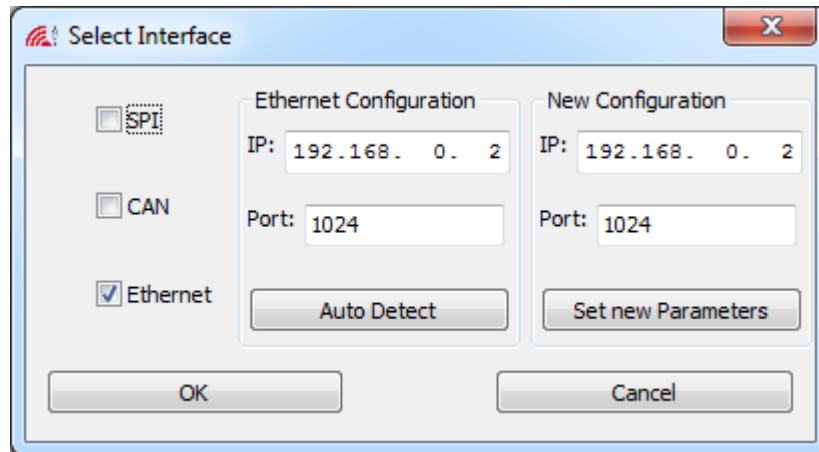


Figure 3.5: Settings of the Ethernet interface

The settings of the Ethernet interface can be configured by selecting the checkbox *Ethernet* of the *Select Interface* dialog (see figure 3.5). In order to connect a radar module via the Ethernet interface (see section 2.5), its IP address and port number must be entered into the corresponding control fields. If these two parameters are unknown, the software can automatically scan the network for available radar modules by pressing the button *Auto Detect*. After successful scan, a dialog with a list of search results appears. Finally, the selected configuration can be accepted by *OK* or rejected by *Cancel*.

Additionally, the connection parameters of a radar module can be modified by entering the new values into the control field *New Configuration* and pressing the button *Set new Parameters*. Hereby, the currently entered IP address and Port number will be adopted and saved into the sensor's EEPROM.

If the connection parameters are unknown and the automatic detection does not work properly, there are two possibilities. The first is to find out the current parameters which are saved in the radar module. The second is to set/reset the parameters. To find out

the saved parameters, these steps should be followed:

- The host computer should be in the same subnet (192.168.0.3 - 192.168.0.255) as the default IP of the radar (see table 2.7).
- The radar module has to be restarted by removing the power supply. After power up, the *AUX* LED is active for 10 seconds. This indicates, that the module can be connected with its default parameters during this time.
- By clicking the *Auto Detect* button within these 10 s, the radar module will be connected with its default parameters and asked for the saved configuration. The dialog that appears, if the process was successful, will show the saved configuration.

To set new parameters without knowing the current parameters, the following steps should be applied:

- The default parameters (see table 2.7) should be configured in the IMST SenTool. To be sure, the values can be entered in the *Ethernet Configuration* and the configuration can be saved by pressing *OK*. If the default parameters are shown in the interface field (see figure 3.2 (a)), they are configured.
- The host computer should be in the same subnet (192.168.0.3 - 192.168.0.255).
- The new parameters have to be entered in *New Configuration*.
- The radar module has to be restarted by removing the power supply.
- By clicking the *Set new Parameters* button, the new parameters can be set within the 10 s time limit.

Note: Each new radar module with an Ethernet interface listens to its default IP address **192.168.0.2** on Port **1024**.

3.1.4 Radar Information

When a radar module is successfully connected to the computer, all functions of the IMST SenTool are activated and become available to the user. In order to access the parameters of the currently connected sensor, the button *Radar Information* of the interface control field can be pressed. Figure 3.6 shows the appearing window which contains several entries with information about the radar's hardware and software settings.

The first category contains the connection parameters which are also visible in the interface control field (see figure 3.2a) after a module has been connected to the computer. As the three interfaces have different parameters, the entries of the *Connection Information* are also different depending on the used interface.

Connection Information

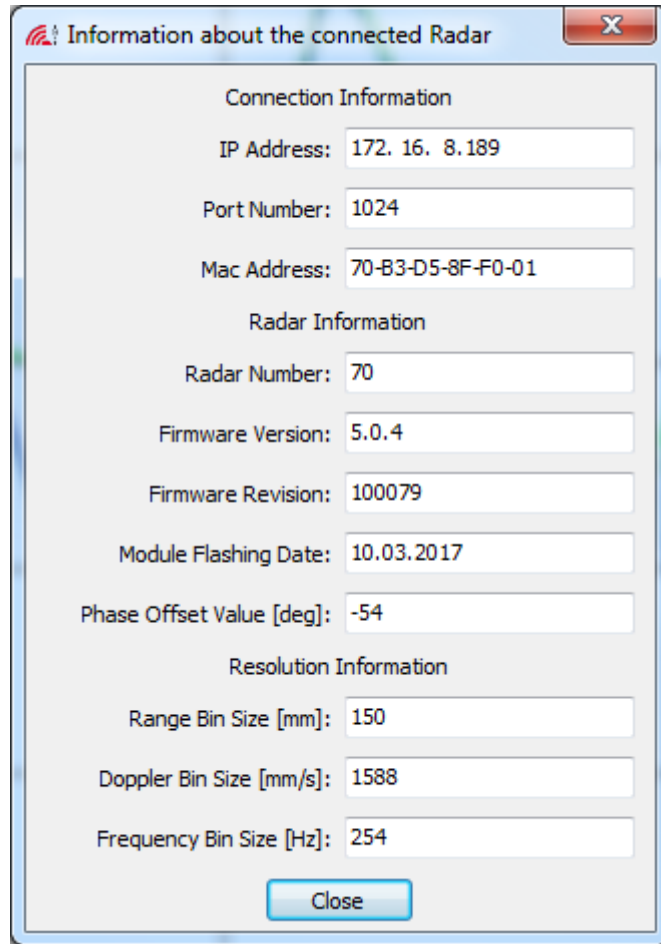


Figure 3.6: Information window for a Radar Module with Ethernet interface

- SPI: For the SPI interface the *Module Name* is displayed.
- CAN: For the CAN Bus interface the *Node ID* and the *Baud Rate* are displayed.
- Ethernet: For the Ethernet interface the *IP Address*, the *Port Number* as well as the *MAC Address* are displayed.

Radar Information

- Radar Number: Unique series number of the connected radar module.
- Firmware Version: The version number of the flashed firmware.
- Firmware Revision: The internal revision number of the flashed firmware. This revision number can be useful for support because different revisions may exist for one firmware version.
- Module Flashing Date: The date when the firmware has been loaded or flashed into the radar module formatted as DDMMYY.

Phase Offset Value [deg]: This offset value in degrees is measured for every radar module and is used for internal correction of phase relations between the two receiving antennas. The phase offset, caused by component tolerances and the manufacturing process, has to be compensated for correct view angle estimation (see section 5.5).

Resolution Information

Range Bin Size [mm]: This value shows the current size of a range bin in [mm] and is actualized at any change of the radar settings.

Doppler Bin Size [mm/s]: This value shows the current size of a doppler bin in [mm/s] and is actualized at any change of the radar settings.

Frequency Bin Size [Hz]: This value shows the current size of a frequency bin in [Hz] and is actualized at any change of the radar settings.

3.2 File Menu

In the upper left corner of the IMST SenTool the *File menu* is located as already shown in figure 3.2 b. The following options are available in the *File* menu.

Interface: This option opens the interface menu which is described in section 3.1. Note that the interface menu is only available in the disconnected mode.

Settings: Here the settings menu can be opened where some options regarding data recording are located (see section 3.10.1).

Hotkeys: Selecting this option opens a dialog showing the available hotkeys as shown in figure 3.7.

Reset Radar: With this option the internal configuration of the radar module can be reset to default values. Note that this option is only available in connected mode.

Exit: Close IMST SenTool.

3.3 Status Bar

When a radar module is connected to the IMST SenTool, the controls become available to the user. Additionally, the status bar at the bottom is filled with different information. Figure 3.8 shows an example of the status bar with a connected sensor during a measurement. The following points describe the fields enumerated in figure 3.8.

1. Here some short information appears when the mouse moves over a control.
2. Shows the temperature on the Frontend board. If the value exceeds 75°C during the measurement, the radar module stops operation.
3. Shows the temperature inside the transceiver chip on the frontend board.
4. This warning sign appears when the currently adjusted frequency is outside the ISM band (24 GHz – 24.25 GHz). Clicking on this sign opens a dialog with further information. Additionally, a warning in the *Message Window* appears.
5. Here the output power in [dBm] of the transceiver chip is shown. It can be adjusted via the *Radar Controls*. This output power is not the emitted power of the radar module. The value also varies over temperature and frequency.
6. This value shows the actual process time in milliseconds during measurements. If no measurement is running, this field is empty. The process time is the total time for starting a measurement, receiving the data and visualizing them. All types of plots provide a control element where a *Time Interval* can be set. If the process time is smaller than this interval, the software waits to satisfy this interval. Otherwise the shown process time turns red and the process works as fast as possible.

3.4 Radar Controls

The *Radar Controls* provide access to the most important parameters for the radar measurement (see figure 3.9). This section contains a description of all *Radar Controls*.

Start/Stop-Frequency:	Here the start- and stop-frequency in [MHz] can be set. The allowed frequency range is defined in section 4. The start frequency must be smaller than the stop frequency. The change of the frequency range must be confirmed by pressing the button <i>Set Frequency</i> . Invalid values will appear red and won't be sent to the radar module.
Set Frequency:	Pressing this button sends the start- and stop-frequency to the radar module provided that the frequencies are valid.
Ramp-Time [ms]:	This value adjusts the time in milliseconds of a FMCW up- or down-ramp. For a CW measurement, it is the time how long the single frequency signal is sampled.
Zero Pad Factor:	Here a factor can be selected by which the time-domain signal is padded with zeros. Possible values are 1, 2, 4 and 8. For values different from 1, the block of time-domain samples of each channel, which always consists of 1024 samples, is subdivided into small blocks with the length of the respective factor. Each small block is averaged resulting in one single

sample and the missing samples are filled with zeros (please note that the averaging also shows a low-pass behavior). By selecting a higher *Zero Pad Factor* the resulting data plot graph of the FFT data is smoothed. The table below shows the relation between factor and the resulting number of samples and zeros.

Zero Pad Factor	Data Samples	Zero Samples
1	1024	0
2	512	512
4	256	768
8	128	896

Attenuation [dB]: Here the output power of the transceiver chip on the Frontend can be reduced by the selected value. As described in section 3.3, the output power varies over temperature and frequency, so the attenuation value may be not exact.

Rx Channel Selection: By enabling/disabling these checkboxes the respective Rx channel is switched ON/OFF where I1/Q1 channels belong to the first receiver antenna and I2/Q2 channels belong to the second receiver antenna. Disabling channels will result in a faster processing time because less data must be processed, transmitted and visualized. Dependent on the active *Tab* the selection may be restricted since some plots need certain channel constellations.

Frontend Power: Here the power supply of the frontend board can be switched ON/OFF. Please note, that the radar module will still receive data when it is switched OFF.

Power Saving: Enables/disables the power saving mode. If the power saving mode is enabled, the frontend power supply is switched ON/OFF at each measurement. This decreases the power consumption as well as the internal sensor temperature. If the power supply is generally switched OFF by the *Frontend Power* control, the ON/OFF switching in power saving mode is disabled.

Normalization: Enables/disables the normalization of FD radar data. If enabled, the measured signal power of a reference target is independent from distance and the selected operation parameters (ramp time, bandwidth, etc.).

3.5 Data Plot

When the IMST SenTool is started, the *Data Plot* tab is selected on default. This plot visualizes the received frequency-domain and time-domain data. In the following section, the characteristics of the *Data Plot* and its controls are described.

3.5.1 Plots

If the *Data Plot* tab is selected, three different data plots can be shown by enabling or disabling the respective checkboxes of *Display Data Type* (see figure 3.10). It is possible to zoom into each plot independently if the checkbox *Zoom* is selected. Therefore, a square has to be drawn into the plot while the left mouse button is pressed. Pressing the right mouse button in the respective plot rescales it. If the checkbox *Marker* is selected, it is possible to place markers on the data graphs. The current value of a marker is displayed in the legend. Pressing the right mouse button on the respective marker deletes it. So each plot can be analyzed independently. These functions work during measurement as well as in stop mode. Taking a screenshot with the *Data Save Controls* (see section 3.10) of a zoomed-in plot is also possible. Leaving the tab, pressing the *Autoscale* button or changing the frequency or the ramp-time will rescale all plots.

3.5.2 Controls

Figure 3.11 shows the controls belonging to this tab. As mentioned above, these controls will only be visible if the tab *Data Plot* is selected by the user.

Start Measurement:	Starts/stops a measurement specified by <i>Measurement Type</i> .
Time Interval [ms]:	Here a time interval in milliseconds between two measurements can be specified. Valid values range from 0 to 1000. As mentioned in section 3.3, the measurement and visualization process will wait to fulfill this interval provided the process time is smaller than the specified value.
Maximum Distance [m]:	This value represents the maximum distance in meters that will be displayed by the plot. Please note that the name of this control changes if another <i>Measurement Type</i> or <i>Unit of the x-Axis</i> is selected. That means that e.g. during CW measurements this control can be used to enter the maximum speed in meters per second. In total, the following possibilities to set the range of the x-axis exist: • distance in meters, • speed in meters per second, • number of Range/Doppler bins,

- intermediate frequency in kHz.

The maximum possible value depends on the selected *Measurement Type* as well as actual radar settings.

Set Distance:	This button sets the value entered in the control element <i>Maximum Distance [m]</i> . The label of the button changes as well dependent on the settings like described above.
Measurement Type:	Here the type of the radar measurement can be selected. With FMCW Up-Ramp, the radar performs a frequency sweep from the start- to the stop-frequency with the a duration specified by the ramp time. With FMCW Down-Ramp, a sweep from the stop- to the start-frequency will be performed. When CW is selected, no sweep will be performed and the start frequency will be sampled for the duration specified by the ramp time.
Display Data Type:	These checkboxes enable or disable different data plots. If more than one plot is activated, the window is split into several sub-plots. Figure 3.10 shows an example where all three plots are activated. All plot combinations are possible and the measurement will stop if no plot is activated. The checkbox <i>FD</i> enables a frequency-domain plot which shows the magnitudes of the FFT data received from the Radar Module. The checkbox <i>Phase</i> enables another frequency-domain plot which shows the phase or the object angle. The checkbox <i>TD</i> enables a time-domain plot that shows the sampled IF signal. Because of the large amount of data, the process time will increase noticeable when the time-domain data plot is activated.
Unit of the x-Axis:	Here the displayed unit of the x-axis of all three plots can be selected. Each selection has two units where the first one belongs to frequency-domain plots (FD, Phase) and the second one to time-domain plot (TD). The possible selections change when the <i>Measurement Type</i> changes. For a FMCW measurement the following selections are possible: • <i>Distance [m] / Time [ms]</i>: Distance in meters for frequency-domain data plots and time in milliseconds for time-domain plot where the range is specified by the control <i>Ramp-Time</i>. • <i>Range Bins / Samples</i>: This selection divides the x-axis into discrete data samples. The time-domain data has always 1024 samples. The number of samples of frequency-domain plots, here called “Range Bins”, depends on the <i>Maximum Distance</i>.

- *Frequency [Hz] / Time [ms]*: This selection sets the unit of frequency-domain data plots to the corresponding IF signal frequency.

When the control *Measurement Type* is set to CW, these selections change into:

- *Speed [m/s] / Time [ms]*: Here the x-axis of frequency-domain data plots corresponds to speed in meters per second.
- *Doppler Bins / Samples*: This selection divides the x-axis into discrete data samples. The number of samples of frequency-domain plots, here called “Doppler Bins”, depends on the maximum target speed.
- *Frequency [Hz] / Time [ms]*: Same as for the FMCW measurements.

Unit of the y-Axis:

The unit of the y-axis can be set to two different options: *[dBm]/Phase [deg]/[V]* and *[dBm]/Obj. Angle [deg]/[V]*.

If the first option is selected, the FD plots show the signal level in dBm (note that the minimum value is limited to -60 dBm) and the signal phase in degrees. The TD plot represents the voltage of the sampled IF signal in Volts.

If the second option is selected, there is one significant difference to the previous one. Here the unit of the FD phase plot changes to Object Angle in degrees. This means that the phase plot shows the object angle of the most dominant scatterer of each range cell. If there is more than one dominant scatterer in the same range cell, the respective value has no meaning. The object angle is measured between the channels I1 and I2 or between the channels Q1 and Q2, so only two data graph can be shown. The range of the object angle is internally limited to -60 and 60 degrees.

Anti Aliasing:

This is a cosmetic setting for the data graphs. If enabled the representation of the graphs gets smoothed but depending on the hardware of the used PC the process time may be increased significantly.

Zoom/Marker:

These checkboxes switch between Zoom and Marker modes. Only one mode can be active at the same time.

Autoscale/Reset Markers:

If the Zoom mode is selected, pressing this button will rescale all plots of the current data sets. In Marker mode all active markers will be deleted.

Max Hold:

This checkbox enables/disables the single channel maximum hold functionality of the *Frequency Domain Data Plot*. A

dashed curve for every activated channel will appear, showing the maximum measured values. To reset these curves this function can be disabled and re-enabled or the *Autoscale* button can be used.

Max Hold All: This checkbox enables/disables the overall maximum hold functionality of the *Frequency Domain Data Plot*. A red dashed curve will appear, showing the maximum measured values of all activated channels. To reset these curves this function can be disabled and re-enabled or the *Autoscale* button can be used.

3.6 Polar Plot

When the *Polar Plot* tab is selected, the GUI shows a half polar coordinate system and the *Polar Plot Controls* appear (see figure 3.12b). This plot is an alternative representation of the frequency-domain data of a FMCW Up-Ramp measurement. It visualizes the magnitudes of dominant scatterers in polar coordinates representing their distances and object angles. The data is shown as circles in different colors representing the receiver channels. The size of a circle indicates the corresponding magnitude, i.e. large circles represent high signal levels. As mentioned before, the object angle is estimated by evaluating the phase difference between two receiver channels (for more details please refer to section 5.5). Hence, both I and/or Q channels must be activated as a pair in the *Rx Channel Selection* in the Radar Controls.

Figure 3.12a shows the controls of this plot which are described below.

Start Measurement:	Starts/stops a radar measurement.
Time Interval [ms]:	Here a time interval in milliseconds between two measurements can be specified. Valid values range from 0 to 1000. As mentioned in section 3.3, the measurement and visualization process will wait to fulfill this interval provided the process time is smaller than the specified value.
Threshold [dBm]:	This value defines a threshold for magnitude data to be shown. Only data greater or equal this threshold will be displayed. Valid values range from -60 to 40. Note that the process time will increase the more data has to be displayed.
Plot Orientation:	Here the orientation of the plot can be adjusted. If the <i>Up</i> option is selected, the plot is oriented from bottom to top, if <i>Right</i> is selected, it is oriented from left to right and so on.
Mirror Plot:	Enables/disables mirroring of the object angle.

Minimum Angle [°]:	This value defines the minimum angle in degrees for data to be shown. Also the plot will be adjusted using this value. The value can be either positive or negative but must be smaller than <i>Maximum Angle</i> .
Maximum Angle [°]:	This value defines the maximum angle in degrees for data to be shown. Also the plot will be adjusted using this value. The value can be either positive or negative but must be greater than <i>Minimum Angle</i> .
Minimum Distance [m]:	This value defines the minimum distance in meters for data to be shown. Also the plot will be adjusted using this value. The value must be smaller than <i>Maximum Distance</i> .
Maximum Distance [m]:	Here the maximum distance in meters can be set. The maximum possible value is dependent on the actual bandwidth. If a value larger than the maximum possible value is set, the value is adjusted automatically. Valid values range from 1 to 1000. In order to transmit a modified value to radar module, the button <i>Set Distance</i> has to be pressed.
Set Parameters:	By pressing this button the adjusted parameters are set and the maximum distance is sent to the radar module provided that the values are valid.
Size Factor:	This slider controls the size of the shown circles. The size scaling also depends on the threshold setting and on the highest actual magnitude level.

3.7 History Plot

By selecting the *History Plot* tab, the *History Plot Controls* and another kind of plot, both shown in figure 3.13, are displayed. Like the *Polar* plot, this is another representation of the frequency-domain data resulting from a FMCW measurement. It shows the signal magnitude in dBm as color-coded intensity values in a distance-over-time plot.

Additionally, a second plot showing the respective phase data can be switched on. Since the x-axis displays the time in seconds, every column of the presented data is associated to a particular time value and represents a whole FMCW radar measurement. This way, many past measurements can be displayed simultaneously as a history. Because data from only one Rx channel can be visualized, the first selected channel from the *Rx Channel Selection* controls is used. The length of the time history and some other parameters can be adjusted in the *History Plot Controls* which are described below.

Start Measurement:	Starts/stops a radar measurement.
--------------------	-----------------------------------

Time Interval [ms]:	Here a time interval in milliseconds between two measurements can be specified. Valid values range from 10 to 1000. As mentioned in section 3.3, the measurement and visualization process will wait to fulfill this interval provided the process time is smaller than the specified interval.
Monitoring Time [s]:	This value sets the maximum monitored time in seconds (corresponding to the history length). Valid values range from 1 to 100. The number of radar measurements simultaneously displayed in the <i>History Plot</i> can be calculated as <i>Monitoring Time / Time Interval</i> . It should be mentioned that a large number of data samples significantly slows down the measurement and visualization process.
Minimum Distance [m]:	This parameter defines the minimum measurement distance in meters. Valid values range from 0 to 999, additionally the <i>Minimum Distance</i> must be smaller than the <i>Maximum Distance</i> .
Maximum Distance [m]:	This parameter defines the maximum measurement distance in meters. The maximum possible value depends on the current radar settings (see section 5.2). If a value larger than the maximum possible value is set, it will be adjusted automatically. Valid values range from 1 to 1000.
Set Parameters:	Pressing this button accepts the changes made in the controls, rescales the axes and clears the <i>History Plot</i> .
Phase:	Enables/disables the <i>Signal Phase</i> plot.
Measurement Type:	Here the type of the radar measurement can be selected. With FMCW Up-Ramp, the radar performs a frequency sweep from the start- to the stop-frequency with the a duration specified by the ramp time. With FMCW Down-Ramp, a sweep from the stop- to the start-frequency will be performed. When CW is selected, no sweep will be performed and the start frequency will be sampled for the duration specified by the ramp time.
Colour Map:	Here another colour representation for the magnitude values can be selected.

3.8 Human Tracker Plot

The tab *Human Tracker* uses the same half polar coordinate system like the *Polar Plot* (see section 3.6) but the data displayed here is not the raw frequency-domain data. Rather the plot is used for visualization of data generated inside the radar module by an algorithm

called Human Tracker. This algorithm can e.g. be applied to detect and track moving targets inside a static background scene, which is typical for many security applications. Figure 3.15 shows a simplified block diagram of this functionality. The algorithm consists of three principal steps:

- suppression of the static background,
- detection of potential targets,
- tracking of the detected targets (optional).

Depending on the parameter *Enable Tracker* in the *Human Tracker Controls*, either a list of detections or a list of tracked targets is shown. In both cases, each target is presented in the polar plot as a dot labeled with its distance in meters, object angle in degrees and magnitude level in dBm. Additionally, the corresponding target ID is shown for tracked targets. Figure 3.14 shows a screenshot the Human Tracker plot and its controls which are described below.

Start Measurement:	Starts and stops a cyclic Human Tracker measurement. This button is initially locked when the <i>Human Tracker</i> tab is selected. Before a measurement can be started, the corresponding parameters must be set. After the button is pressed, an initial calibration is performed and the process is started.
Time Interval [ms]:	Here a time interval in milliseconds between two measurements can be specified. Valid values range from 10 to 1000. As mentioned in section 3.3, the radar control of the GUI will wait if the measurement and visualization time is smaller than the specified interval.
Minimum Angle [°]:	This value defines the minimum angle in degrees for data to be shown. Also the plot will be adjusted using this value. The value can be either positive or negative but must be smaller than <i>Maximum Angle</i> .
Maximum Angle [°]:	This value defines the maximum angle in degrees for data to be shown. Also the plot will be adjusted using this value. The value can be either positive or negative but must be greater than <i>Minimum Angle</i> .
Minimum Distance [m]:	This value defines the minimum distance in meters for data to be shown. Also the plot will be adjusted using this value. The value must be smaller than <i>Maximum Distance</i> .
Maximum Distance [m]:	Here the maximum distance in meters can be set. The maximum possible value depends on the actual bandwidth (refer to section 5.2). In case of an invalid input value, the control is automatically adjusted to the maximum possible range.

Minimum Distance [m]:	Here the minimum distance in meters can be set. Valid values range between 0 m and <i>Maximum Distance</i> . In case of an invalid input value, the control is automatically adjusted to the last valid range.
Parameters:	This control element allows a quick selection of all parameters used to control and configure the Human Tracker algorithm. After a parameter has been selected in the drop-down list, the corresponding value can be changed in the field below.
<i>Reference Pulses:</i>	This value defines the total number of calibration measurements performed during the initialization process of the Human Tracker algorithm. Valid values range from 1 to 1000.
<i>Number of Targets:</i>	This value determines how many targets can be simultaneously processed by the Human Tracker algorithm. The radar looks for potential targets between <i>Minimum Distance</i> and <i>Maximum Distance</i> or, alternatively, if the maximum number of targets have been detected. Valid values range from 1 to 10.
<i>Background Filter Factor:</i>	This value defines how fast the background signal stored in the radar module is adopted to a dynamically changing scene (see figure 3.15) and corresponds to N in equation 3.2. Valid values range from 1 to 1000.
<i>Noise Filter Factor:</i>	This value defines how fast the detection threshold is adopted to a dynamically changing noise power (see figure 3.15) and corresponds to M in equation 3.3. Valid values range from 1 to 1000.
<i>Over Threshold Factor:</i>	This value defines the factor by which the amplitude of a valid target must exceed the detection threshold. To avoid floating-point numbers, the unit is $[1/10]$, i.e. the value 30 corresponds to a factor of 3.0. Valid values range from 10 to 1000.
<i>Minimum Separation [bins]:</i>	This value determines the minimum spatial separation between two detected targets in resolution bins. This way, a single spatially distributed target will not be interpreted as two or more different targets. Valid values range from 1 to 512.
<i>Maximum Range Shift [m]:</i>	This value determines the assumed maximum displacement of a target between two measurements. During the tracking process, a detected target is only assigned to an existing track if it's position offset is smaller than

Maximum Range Shift. Due to the limited measurement accuracy of the system, this parameter mainly depends on the maximum expected position error rather than on the dynamic properties of the target. Valid values range from 0 to 1000.

Maximum Detection Losses: If a target cannot be confirmed during the tracking process, a corresponding *Loss Counter* is increased by one. If this counter is greater than the parameter defined by *Maximum Detection Losses*, the track is deleted from the Human Tracker database. Valid values range from 0 to 100.

Tracker Enabled: Enables/disables the tracking functionality of the Human Tracker algorithm in the radar module. If disabled, the radar will start transmitting a list of detections instead of a list of tracks. To accept this setting, the button *Set New Parameters* has to be pressed.

Set Parameters: Pressing this button accepts the adjusted parameters provided the values are valid. If Human Tracker parameters were changed, the radar performs a number of FMCW measurements, specified by the parameter *Reference Pulses*, to calculate a background signal for clutter suppression. These measurements are performed in time steps specified by the parameter *Time Interval*. Consequently, the radar module will need $\text{Reference Pulses} \times \text{Time Interval}$ milliseconds for the initial calibration process. This time is also shown in the first entry of the *Status Bar*.

Show IDs: Enables/disables displaying of the ID number of the detections/tracks.

Show Labels: Enables/disables displaying a label for each detection/track showing the corresponding distance in meters, the angle in degrees and the signal level in dBm.

Plot Orientation: Here the orientation of the plot can be adjusted. If the *Up* option is selected, the plot is oriented from bottom to top, if *Right* is selected, it is oriented from left to right and so on.

Mirror Plot: Enables/disables mirroring of the object angle.

In following, the processing steps of the Human Tracker algorithm are briefly described. In order to achieve a sufficient sensitivity to detect low reflective targets (e.g. humans) inside a high reflective background (e.g. buildings, parking cars, fences), the static background (also known as clutter) must be removed from the measurement data. This is done

independently for both Rx channels directly after sampling the IF signals. In a first step, the differential signal S_D is calculated using the formula:

$$S_D(t) = S_M(t) - C(t), \quad (3.1)$$

where S_M is the sampled IF signal in time domain and C is the corresponding background signal. In a second step, the background (clutter) information is updated:

$$C^*(t) = \frac{C(t) \cdot (N - 1) + S_M(t)}{N}. \quad (3.2)$$

The parameter N determines how fast the background data stored in the radar module adopts to the measurement data (refer to control *Background Filter Factor*).

After a successful clutter suppression the radar data contains only contributions coming from noise and moving targets. At this point, the data is transformed to frequency domain using a Fast Fourier Transform (FFT) of order 1024. After comparison of the signal spectrum $A(f)$ with a corresponding detection threshold $D(f)$, a number of potential targets can be identified. The detection threshold describes the expected noise level of the signal for each frequency respective range bin. As this value is not constant over time, it must be updated continuously during the measurement process:

$$D^*(f) = \begin{cases} \frac{D(f) \cdot (M-1) + A(f)}{M} & \text{if } A(f) < D(f) \cdot k \\ D(f) & \text{else} \end{cases} \quad (3.3)$$

Hereby, M is a filter constant (analog to N in equation 3.2) and k is a factor by which a valid detection must exceed the detection threshold (refer to controls *Noise Filter Factor* and *Over Threshold Factor*). It should also be mentioned that the threshold is not updated at the positions of detected targets. Additionally, the new list of detections is passed to a tracker algorithm, which tries to assign them to a target list stored in the radar module memory. Hereby, three different cases can appear:

1. an existing target is confirmed,
2. a new target is found and added to list,
3. an old target is lost and (optionally) deleted.

The output data stream from a single radar module to the central control unit contains a list of detected and tracked targets with the following properties:

- unique target identifier,
- radial distance from radar,
- view angle relative to radar line-of-sight,
- signal amplitude,
- life-time (how long is the target visible),
- status flag (new or confirmed target).

Hotkeys	
<i>General Controls</i>	
ESC	Quit Program
SPACE	Start/Stop Measurement
PAGE UP	Switch Plot Tab forward
PAGE DOWN	Switch Plot Tab backward
F1	Switch to Data Plot
F2	Switch to Polar Plot
F3	Switch to History Plot
F4	Switch to Human Tracker Plot
F5	Switch to Range Finder Plot
ARROW UP	Scroll controls upwards
ARROW DOWN	Scroll controls downwards
F11	Scroll information text upwards
F12	Scroll information text downwards
ALT+1	Show/Hide FD plot (Data Plot) or Distance plot (Range Finder)
ALT+2	Show/Hide Phase plot (Data Plot, History Plot) or Velocity plot (Range Finder)
ALT+3	Show/Hide TD plot (Data Plot) or Angle plot (Range Finder)
ALT+A	Autoscale/Reset Markers in Data Plot
ALT+D	Start/Stop saving Radar Data
ALT+F	Open File menu for further options
ALT+L	Load file when disconnected
ALT+M	Maximize/Minimize Window
ALT+P	Save Picture of current plot
<i>Radar Controls</i>	
ALT+B	Set Frequencies
ALT+C	Connect/Disconnect a Radar
ALT+I	Show Radar Information
ALT+R	Reset Range Finder
ALT+S	Set Parameters dependent on active plot
<i>Replay Mode</i>	
SPACE	Start/Stop playing
ARROW LEFT	Previous Measurement
ARROW RIGHT	Next Measurement
END	Stop playing and Reset

Figure 3.7: Hotkeys



Figure 3.8: Status bar at the bottom of the GUI

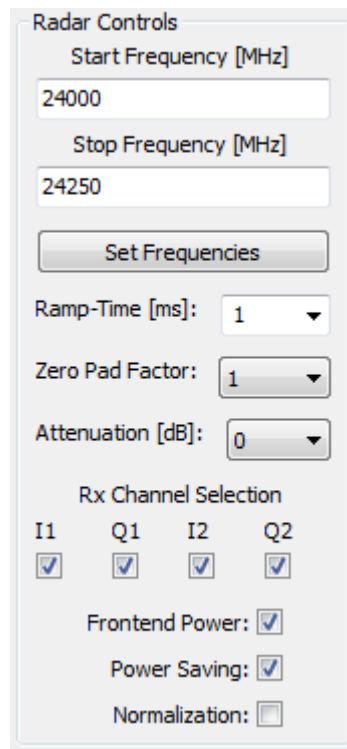


Figure 3.9: Radar Controls

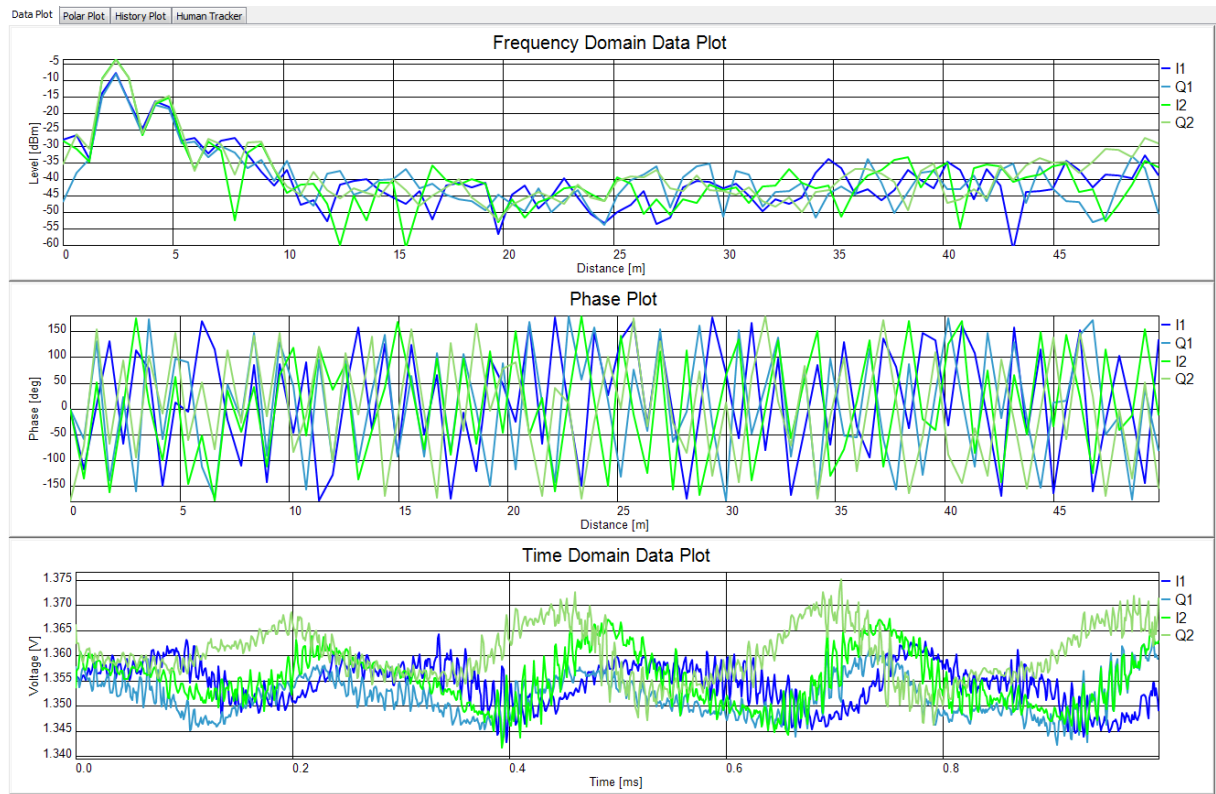


Figure 3.10: Graphs of the Data Plot tab

The 'Data Plot Controls' panel includes the following settings:

- Start Measurement**: Button
- Time Interval [ms]**: 50
- Max. Distance [m]**: 10
- Set Distance**: Button
- Measurement Type**: FMCW Up-Ramp (dropdown)
- Display Data Type**: FD ☒ Phase ☐ TD ☐
- Unit of the x-Axis**: Distance [m] / Time [ms] (dropdown)
- Unit of the y-Axis**: [dBm] / Phase [deg] / [V] (dropdown)
- Anti Aliasing**: ☐
- Zoom**: ☒ **Marker**: ☐
- Autoscale**: Button
- Max Hold**: ☐ **Max Hold All**: ☐

Figure 3.11: Data Plot Controls

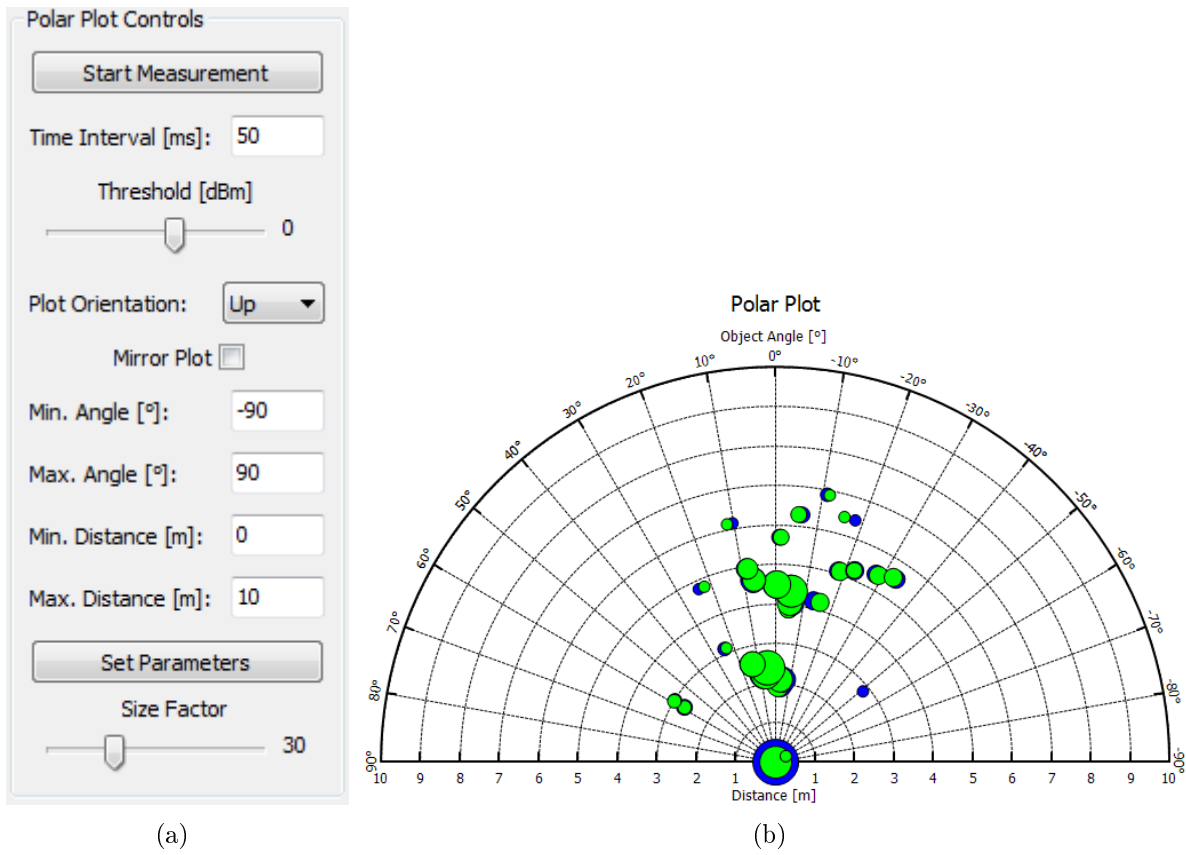


Figure 3.12: (a) Polar Plot controls, (b) Polar Plot graph

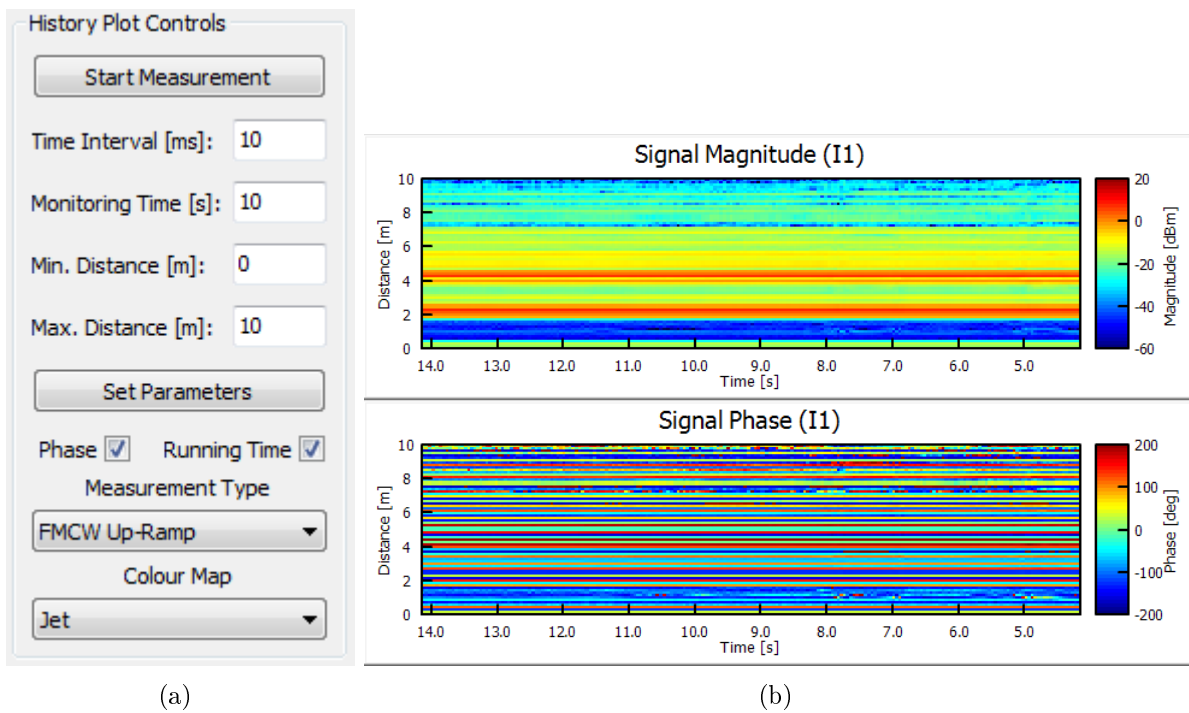
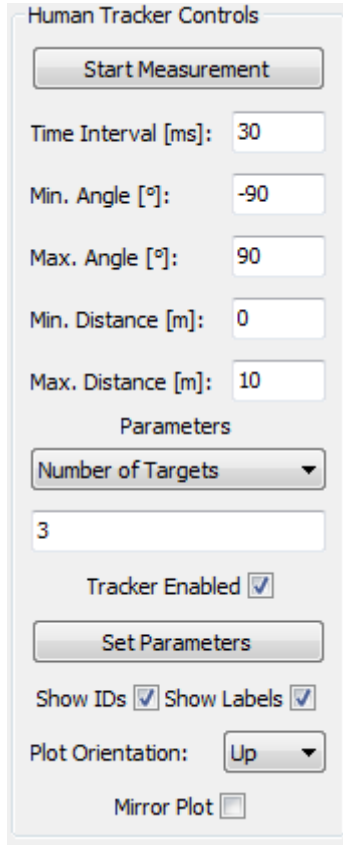
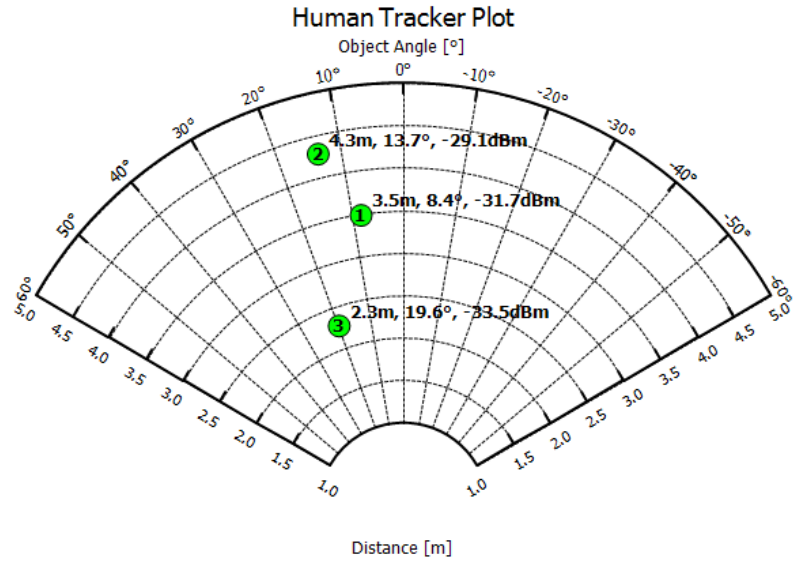


Figure 3.13: (a) History Plot controls, (b) History Plot graph



(a)



(b)

Figure 3.14: (a) Human Tracker controls, (b) Human Tracker graph

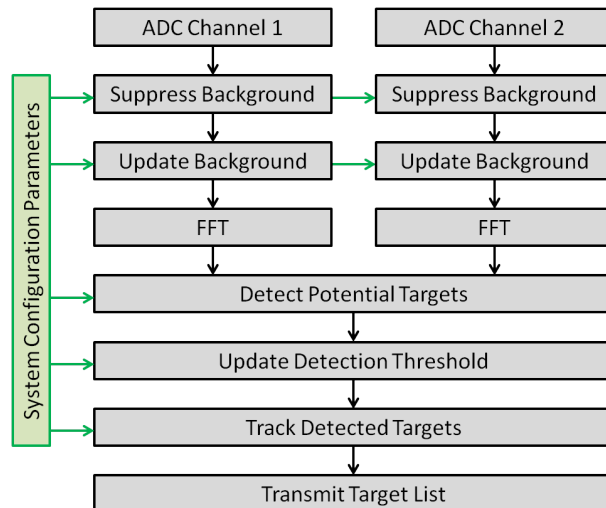


Figure 3.15: Block diagram of the Human Tracker algorithm (the green arrows indicate configuration parameters of the corresponding functional blocks)

3.9 Range Finder Plot

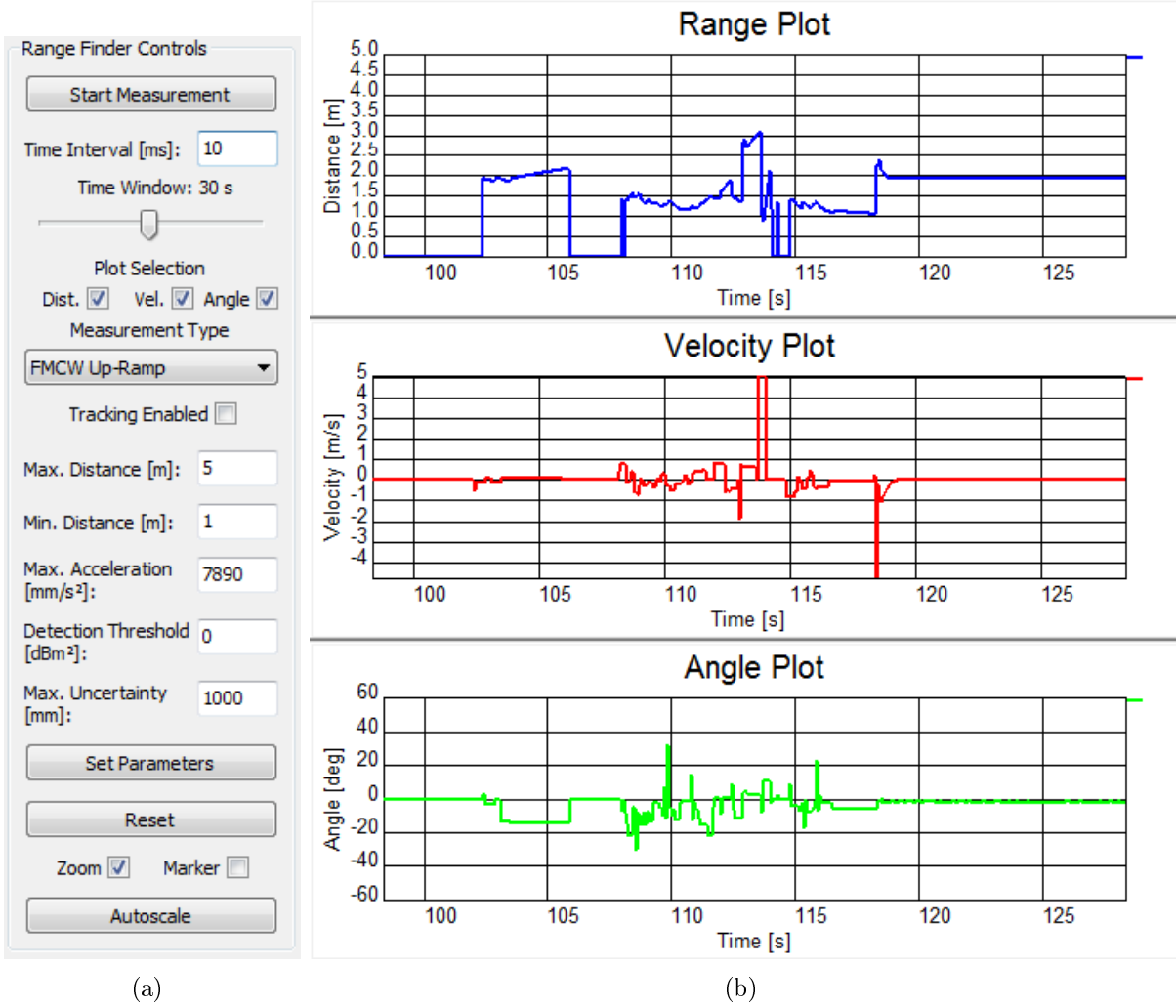


Figure 3.16: (a) Range Finder controls, (b) Range Finder graph

The tab *Range Finder* of the IMST SenTool visualizes measurement data generated by the Range Finder algorithm implemented in firmware of the sentireTM Radar Module (for user command description refer to /CMD/). This function can be used to estimate the distance and the radial velocity of a potentially moving target. Additionally, the object angle of the detected target can also be monitored (see figure 3.16b).

In order to measure range and speed of a target simultaneously, a linear Kalman filter is implemented in the Range Finder algorithm. It resolves the Range-Doppler-Ambiguity (see section 5.4) caused by moving targets in FMCW radar mode, which allows accurate range measurements even at high radial velocities. Moreover, short time intervals in which the target is not “visible” can be bypassed by the tracking algorithm. Thus, a continuous range (and velocity) history of a target becomes available.

Figure 3.16a shows all control elements appearing in the *Control Panel* if the *Range Finder* tab is activated. These controls are described below.

Start Measurement:	Starts and stops a cyclic Range Finder measurement based on the specified parameters.
Time Interval [ms]:	Here a time interval in milliseconds between two measurements can be specified. Valid values range from 10 to 1000. As mentioned in section 3.3, the radar control of the GUI will wait if the measurement and visualization time is smaller than the specified interval.
Time Window:	This control specifies the maximum time period in seconds visualized in the plots. If the duration of a measurement cycle exceeds this value, the time axis is shifted by discarding the older data points. Valid values range from 1 to 60.
Plot Selection:	This control contains three checkboxes for enabling/disabling the different data plots.
Measurement Type:	Here the direction of the transmitted frequency ramp can be specified. Possible options are Up-Ramp and Down-Ramp. Please note that the behavior of the Kalman filter is slightly different for different ramp types.
Tracking Enabled:	This checkbox enables/disables the tracking functionality of the Range Finder algorithm. If disabled, the raw range measurement values are presented and a velocity estimation is not possible.
Maximum Distance [m]:	Here the maximum distance in meters can be set. The maximum possible value depends on the actual bandwidth (refer to section 5.2). In case of an invalid input value, the control is automatically adjusted to the maximum possible range.
Minimum Distance [m]:	Here the minimum distance in meters can be set. Valid values range between 0 m and <i>Maximum Distance</i> . In case of an invalid input value, the control is automatically adjusted to the last valid range.
Maximum Acceleration:	This parameter defines the maximum possible acceleration in [mm/s ²] of the expected target and, in such way, influenced the dynamics of the Kalman filter.
Detection Threshold:	This parameter defines the Radar-Cross-Section (RCS) of the expected target. The Range Finder algorithm always activates the <i>Normalization</i> feature of the radar system and sets the <i>Zero Pad Factor</i> to 1 (see section 3.4). Consequently, the signal power resulting from a particular target is (roughly) independent from its distance and the corresponding IF. This gives the user the possibility to define a <i>constant</i> detection threshold to distinguish between targets and noise. The algorithm selects the closest target with an RCS above the specified threshold.

Maximum Uncertainty:	This parameter defines the standard deviation of the range estimation in [mm], at which the Kalman filter discards the current measurement values and initiates an internal reset. If a large value is selected, the filter will do many prediction steps (even if no measurement data is available) before this limit is reached. If a small value is selected, the filtered data will be more reliable but can possibly contain “gaps” if no target could be found. The <i>Maximum Uncertainty</i> should be larger than the size of a resolution bin.
Set Parameters:	After changing some configuration parameters, this button must be pressed in order to load the new parameters into the radar module.
Reset:	This button resets the internal state of the internal tracking filter. Hereby, all target information from the past is lost.
Zoom/Marker, Autoscale,	
Reset Markers:	The behavior of these controls is described in section 3.5.2 on page 26.

Note: Before starting the Range Finder algorithm, it is recommended to check the magnitude of Frequency-Domain data using the regular *Data Plot* tab. It is up to the user to ensure that the noise power is lower than the selected *Detection Threshold* over the whole measurement range. Additionally, it is useful to check the true signal power resulting from the expected target (if possible). It is important to understand clearly that the Range Finder algorithm has no *adaptive* behavior like the Human Tracker (see section 3.8), so the measurement parameters should be chosen carefully.

3.10 Data Saving

The IMST SenTool features different functions to save the data measured by the sentire™ Radar Module. This functionality can be called using the *Data Save Controls* explained in section 3.10.2. Anyway, some attention should be paid on the things that happen in the background. The following section describes the configuration of the Data Save Functions, in order to save different kind of data and to understand the saved data files.

3.10.1 Settings Dialog

Figure 3.17 shows the *Settings* dialog which can be entered via File → Settings. In this menu, the data format and the location of the saved measurement data can be configured.

Screenshots

- | | |
|--------------------|---|
| Save Directory: | Specifies the directory for saving screenshots. By default, a folder named <i>Saved Plots</i> is created in the source folder of the software even if the path is deleted in the control. |
| Screenshot Format: | Here the image format of the screenshots from different plots is specified. The following choices are possible: <ul style="list-style-type: none">• Portable Networks Graphics (*.png),• JPEG (*.jpg),• Windows Bitmap (*.bmp). |

Radar Data

- | | |
|------------------|--|
| Save Directory: | Specifies the directory for saving radar data. By default, a folder named <i>Saved Radar Data</i> is created in the source folder of the software even if the path is deleted in the control. |
| Raw Data Format: | This control specifies the file format of raw data measured by the radar module. There are the two principal options: <i>Text Format</i> and <i>Binary Format</i> . It is also possible to select both at once or just one of them. Section 3.10.3 contains a description of both data formats. Please note that saving raw measurement data can only be accessed from the <i>Data Plot</i> tab. |

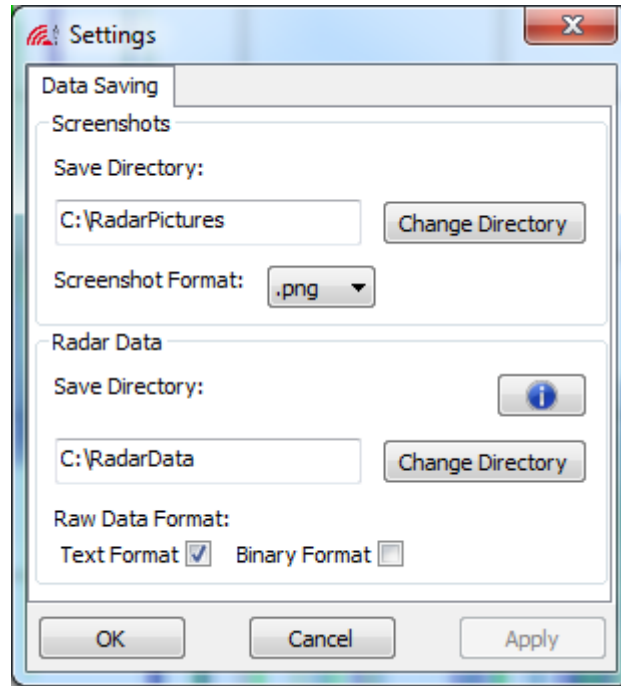


Figure 3.17: Settings dialog

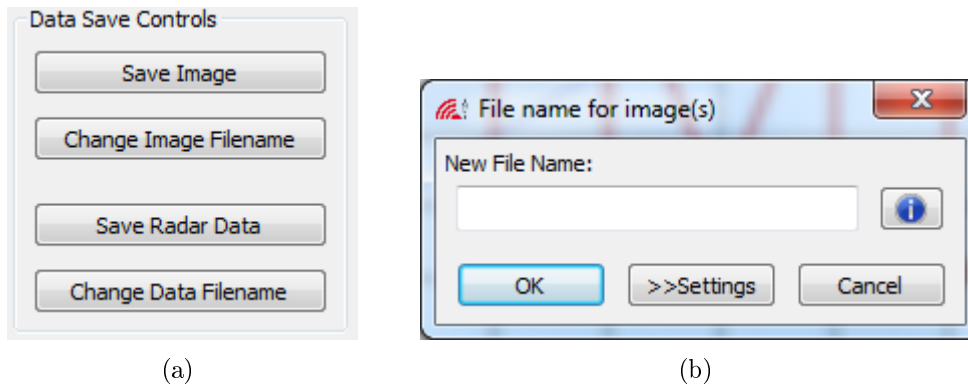


Figure 3.18: (a) Data Save controls, (b) File Name dialog

3.10.2 Data Save Controls

The *Data Save Controls*, shown in figure 3.18a, consist of two buttons for images and two buttons for raw radar data. The dialog window for changing the file name (see figure 3.18b) can be opened by pressing the buttons *Change Image File Name* and *Change Data File Name*. Additionally, a link to the *Settings* dialog window (see Section 3.10.1) is provided. If a file name is not specified by the user, a default name will be generated for every new file. More information relating file names can be found in Section 3.10.3.

Saving measurement data as images follows the principle “What you see is what you get”. That means that an image of a plot, saved by pressing the *Save Image* button, has exactly the same size like in the corresponding plot. If more than one plot is active, an image for each active plot is generated and saved. If the *FD*, *Phase* and *TD* plots are activated in the *Data Plot* tab, an image for all of them is saved. If the zoom function is used, the

corresponding zoomed view is saved. Saving images works during a measurement but also in stop-mode.

Saving Radar Data in the Data Plot

In the *Data Plot* it is possible to save radar data during a measurement and in stop-mode. To save data in stop-mode, at least one measurement must have been performed before (i.e. the plot shows measurement data). If the *Save Radar Data* button is pressed during a measurement, the GUI begins to save all measured data that is shown in the active plots in the format specified in the *Settings* dialog. Therefore, all controls that could influence the active measurement will be locked while data saving is active. To stop the data saving process, the button has to be pressed again. Alternatively, it is stopped when the Radar is disconnected or another tab of the GUI is selected.

The data format of measurements saved from *Data Plot* depends on the currently active plots as well as on the settings *Measurement Type Selection* and *Unit of the y-Axis*. It is differentiated between saving frequency- and time-domain data because a separate file is generated for each domain. This means that the magnitude and phase (respective object angle) data of each radar measurement is saved to a frequency-domain data file. The same is applied to the samples of the time-domain data.

If the text file format is selected by the user, a new `*.txt` file for every measurement of the Radar Module is generated. In case of a binary format, a single file is created for each domain and continuously filled with measurement data while saving is active. If the size of a binary file exceeds 256 MB, a new file is created.

Saving Data in the Polar Plot

In the *Polar Plot* tab, the *Save Radar Data* button becomes available when a measurement is active. Like for the *Data Plot* tab, the most controls are locked while saving data is active. Since the *Polar Plot* shows just another representation of the frequency-domain data measured by an Up-Ramp FMCW measurement, the data saved here consists of magnitude and object angle for the I and/or Q channels.

If the text file format is selected by the user, a new `*.txt` file for every measurement of the Radar Module is generated. In case of a binary format, a single file is created and continuously filled with measurement data while saving is active. If the size of a binary file exceeds 256 MB, a new file is created.

Saving Data in the History Plot

Saving data in the *History Plot* tab works in the same way as for the *Polar Plot*. The button becomes only available when a measurement is active. The *History Plot* shows magnitude and phase information of frequency-domain measurements over time, so the data which is saved here consists of magnitude and phase data dependent on the specified measurement type.

Here the same format and size limitations as for the *Data Plot* or *Polar Plot* can be applied.

Saving Data of the Human Tracker

Saving data in the *Human Tracker* tab works in the same way as for the *Polar Plot* or *History Plot*.

If the text file format is selected, a single *.txt file is generated to save every measurement of the Human Tracker. If the size of the text file exceeds 5 MB, a new file is created. In case of the binary format, also a single file is created. If the size of a binary file exceeds 256 MB, a new file is created.

Saving Data of the Range Finder

Saving data in the *Range Finder* tab works in the same way as for the *Human Tracker*. The button becomes only available when a measurement is active. It is also possible to save data in text and/or binary format with the same size limitations.

3.10.3 File Structure

The data measured by the Radar Module can be saved in text format (see section 3.10.4) as well as in binary format (see section 3.10.5). This section contains some conventions relating the file names and directories of the saved radar measurement data.

Default File Names

As mentioned before, default file names will be used if no user-defined name is specified. This default name consists of a prefix representing the data type combined with the current date and time. Consequently, the default name of a frequency-domain text file will look like:

FD-Data_YYYY-MM-DD_hh-mm-ss.ms.txt.

The same principle is applied to the standard file names of saved images, e.g.:

FD-Data_YYYY-MM-DD_hh-mm-ss.ms.png.

If a file name has been specified by the user, the default prefix is replaced by this name but the date and time information is still appended. This behavior makes it is very unlikely to overwrite an existing file by mistake. Consequently, by entering the file name *UserFilename*, the corresponding name of a binary file will look like:

UserFilename_YYYY-MM-DD_hh-mm-ss.ms.bin.

If the maximum file size has been reached (5 MB for text files, 256 MB for binary files), a new file is created. The name of the new file will be the same as the name of the previous

one but a counting number is appended to it like *_2, *_3 and so on.

Directories

In the *Settings* dialog (see figure 3.17), a directory for the saved images and Radar data can be specified by the user. By default, the standard directory is used, which is the source path of the IMST SenTool. However, various sub-folders are additionally created in the specified directory depending on the requested type of radar data. In contrary, saved images are directly stored in the specified folder.

If no measurement process is running in the *Data Plot*, the last set of measurement data can still be saved to files. In this case, a sub-folder named TD is created for time-domain data and a sub-folder named FD is created for frequency-domain data.

For Radar data saved from the *Data Plot*, *Polar Plot* or *History Plot* during a running measurement, a sub-folder named in the format YYYY-MM-DD_hh-mm-ss.ms will be created to indicate a new measurement session. This session folder contains the same sub-folders as described for the stop-mode (FD and TD).

In case of the *Human Tracker* plot (see section 3.8) or the *Range Finder* plot (see section 3.9), the additional sub-folder named HT resp. RF will be created in the specified (or default) directory. The principle of different directories helps the user to assign the saved data to a particular measurement, even if the file names don't indicate the type of data contained in the file.

3.10.4 Format of Text Files

If a set of radar data is saved in text format, the results of each single frequency ramp (or CW measurement) are stored to a new ASCII data file, whose name convention is specified in section 3.10.3. Each text file consists of a file header (see table 3.1) followed by a matrix of data samples. Each row represents a time sample in TD or a frequency sample in FD which is equivalent to range or doppler bins (see sections 5.2 and 5.3). The data samples in each row are separated by tabs. The text files are readable using any text editor program, so the user can access the measurement data as well as the corresponding radar configuration parameters without the need of additional software tools.

The main disadvantage of saving measurement data to text files is the large resulting number of files as well as increased requirements regarding the HDD storage place. Moreover, the combination of several files to one data set may be rather complicated. In order to save and process large amounts of radar data, the binary format is explicitly recommended (see section 3.10.5).

The results of a Human Tracker measurement (see section 3.8) are also saved to an ASCII text file if the text format is selected. Each HT file starts with a header which is similar to the measurement data header but also contains additional settings of the Human tracker algorithm (see table 3.2). In contrary to the raw measurement data, each file also contains a list of HT target detections (or tracks) rather than the results from

Parameter	Description
Date	format: YYYY-MM-DD
Time	format: hh:mm:ss:ms
Radar Number	unique identifier of the radar module
Interface	SPI, CAN or Ethernet
Start-Frequency [MHz]	or Tx frequency in CW mode
Stop-Frequency [MHz]	obsolete in CW mode
Ramp Time [ms]	or pulse time for CW mode
Attenuation [dB]	Tx attenuation during measurement
Bin Size [mm] / [mm/s]	depending on Radar mode (FMCW or CW)
Number of Samples	TD: 1024, FD: number of range bins
Zero Pad Factor	1, 2, 4, or 8
Normalization	0 or 1 (valid for FD data only)
Active Channels	any combination of I1, Q1, I2, Q2
Type of saved data	FD or TD

Table 3.1: Header of FD and TD text files

a single measurement. Each detected or tracked target is characterized by a number of parameters listed in table 3.3.

The result of each Human Tracker measurement contains a group of several simultaneously detected targets. Each target group is characterized by a time stamp representing the machine time of the computer running the IMST SenTool software. After the header, a HT text file contains a flexible number of data blocks, consisting of a time stamp, the target number in the current HT sequence and a list of target parameters corresponding to table 3.3. Table 3.4 shows an example of Human Tracker measurement data.

The results of the Range Finder measurement (see section 3.9) are also stored in a ASCII data file if the text format is selected. Analog to HT text files, each RF file starts with a header (see table 3.10.4). All following measurements are written after the header, until saving or measurement process is stopped or the file size exceeds 5 MB. Table 3.6 shows an example how the RF measurement data is stored. Each measurement consists of a timestamp which represents the current Radar time in the Epoch Unix time in seconds, the measured range in meters, the corresponding angle in degrees and the estimated velocity in meters per second.

Parameter	Description
Date	format: YYYY-MM-DD
Time	format: hh:mm:ss:ms
Radar Number	unique identifier of the radar module
Interface	SPI, CAN or Ethernet
Start-Frequency [MHz]	start FMCW frequency
Stop-Frequency [MHz]	stop FMCW frequency
Ramp Time [ms]	FMCW ramp duration
Attenuation [dB]	Tx attenuation during measurement
Bin Size [mm]	Size of a range bin
Number of Samples	number of range bins
Threshold Filter Factor	see section 3.8
Over Threshold Factor	see section 3.8
Minimum Target Separation [Bins]	see section 3.8
Minimum Target Distance [m]	see section 3.8
Maximum Detection Losses	see section 3.8
Maximum Range Shift [m]	see section 3.8
Enable Tracker	see section 3.8
Maximum Number of Targets	see section 3.8
Background Filter Factor	see section 3.8

Table 3.2: Header of a Human Tracker text file

Parameter	Description
Target ID	unique target identifier
Target tracked	0: new target, 1: confirmed (tracked) target
Detection counter	lifetime of a target (if Tracking is enabled)
Level [dBm]	signal level of a particular target
Distance [m]	ground distance between radar and target
Angle[deg]	object angle of a particular target (positive: left)

Table 3.3: Target parameters stored in a Human Tracker text file

Time: hh:mm:ss:ms
No. of Targets: m
Target ID: n
Target tracked: 0 or 1
Detection counter:
Level [dBm]:
Distance [m]:
Angle [deg]:
Target ID: n+1
Target tracked: 0 or 1
Detection counter:
Level [dBm]:
Distance [m]:
Angle [deg]:
Target ID: m
Target tracked: 0 or 1
Detection counter:
Level [dBm]:
Distance [m]:
Angle [deg]:
Time: hh:mm:ss:ms
No. of Targets: m
Target ID: n
Target tracked: 0 or 1
Detection counter:
...

Table 3.4: Sequence of Human Tracker data in the text file

Parameter	Description
Date	format: YYYY-MM-DD
Time	format: hh:mm:ss:ms
Radar Number	unique identifier of the radar module
Interface	SPI, CAN or Ethernet
Start-Frequency [MHz]	start FMCW frequency
Stop-Frequency [MHz]	stop FMCW frequency
Ramp Time [ms]	FMCW ramp duration
Attenuation [dB]	Tx attenuation during measurement
Bin Size [mm]	Size of a range bin
Number of Samples	number of range bins
Measurement Type	Up-Ramp or Down-Ramp
Maximum Acceleration	see section 3.9
Minimum Distance	see section 3.9
Detection Threshold	see section 3.9
Maximum Uncertainty	see section 3.9
Tracking Enabled	0: disabled, 1: enabled

Table 3.5: Header of a Range Finder text file

Radar Time [s]
Range [m]
Angle [deg]
Velocity [m/s]

Radar Time [s]
Range [m]
Angle [deg]
Velocity [m/s]

...

Table 3.6: Sequence of Range Finder data in the text file

3.10.5 Format of Binary Files

The sentireTM Radar Module is able to generate a huge amount of data during a relatively short time interval. In order to handle this data, it is recommended to store the measurement data in a binary file. In analogy to text files, each binary data file starts with a header containing the most relevant settings of the radar module during measurement and some algorithm specific parameters if the *Human Tracker* or *Range Finder* is activated (see tables 3.7, 3.8, 3.9). The data stream following the file header is defined by the single bits of the parameter *Data Format*.

Binary Files saved from Data, Polar and History Plot

All samples of the measurement data are written to file in 32-bit floating-point format corresponding to the IEEE-754 standard.

The order of data samples is given by the following rule:

1. a file contains several records (i.e. radar pulses),
2. a record contains several channels (e.g. I1, Q1, I2, Q2),
3. a channel contains several samples,
4. a sample contains one or two sub-samples (e.g. magnitude and phase).

Additionally, each record is initiated with a time-stamp of the corresponding radar measurement. The time is given as a 64-bit unsigned integer value and corresponds to the Epoch Unix Time in milliseconds (time since January 1st 1970).

Example

In the first step, the header data must be extracted from file (see table 3.7). We assume that the file header contains the following information:

- *Data Format* = 12 → FMCW mode, FD data, magnitude and phase,
- *Number of Samples* = 100 → each channel contains 100 range bins,
- *Active Channels* = 10 → channels Q1 and Q2 are active.

Now, we can read the rest of the binary data stream:

while not end-of-file:

read measurement time-stamp in [ms]

for (channel = 1 to 2):

for (sample = 1 to 100):

read magnitude value in [dB]

read phase value in [deg]

Parameter	Data Type	Description
File creation Time [ms]	uint64	Epoch Unix time format
Radar Number	uint32	unique identifier of the radar module
Start-Frequency [MHz]	uint16	FMCW: start frequency, CW: Tx frequency
Stop-Frequency [MHz]	uint16	FMCW stop frequency, CW: not used
Ramp Time [ms]	uint16	FMCW: ramp duration, CW: sampling time
Attenuation [dB]	float32	Tx attenuation during measurement
Data Format	uint32	Binary mask of the stored measurement data: • Bit 0 (LSB): 0 $\rightarrow$ FMCW, 1 $\rightarrow$ CW • Bit 1: TD data in [V] • Bit 2: Magnitude of frequency-domain (FD) data in [dB] • Bit 3: Phase of FD data in [deg] • Bit 4: Object angle in [deg] • Bit 5: Human Tracker data • Bit 6: Range Finder data Arbitrary combination of bits 2, 3 and 4 are possible, e.g. Bit 2 and Bit 3 are set: stored data samples contain magnitude and phase of the complex FD signal.
Bin Size [mm] / [mm/s]	float32	FMCW: range resolution CW: velocity resolution
Number of Samples	uint16	TD: 1024, FD: number of range bins
Bin Size [Hz]	uint32	frequency resolution
Zero Pad Factor	uint16	1, 2, 4 or 8
Normalization	uint16	0 or 1 (valid for FD data only)
Active Channels	uint16	Binary mask of active radar channels: • Bit 0 (LSB): I1 • Bit 1: Q1 • Bit 2: I2 • Bit 3: Q2

Table 3.7: Header of FD and TD binary files

Parameter	Data Type	Description
File creation Time [ms]	uint64	Epoch Unix time format
Radar Number	uint32	unique identifier of the radar module
Start-Frequency [MHz]	uint16	FMCW: start frequency, CW: Tx frequency
Stop-Frequency [MHz]	uint16	FMCW stop frequency, CW: not used
Ramp Time [ms]	uint16	FMCW: ramp duration, CW: sampling time
Attenuation [dB]	float32	Tx attenuation during measurement
Data Format	uint32	Binary mask of the stored measurement data: • Bit 0 (LSB): 0 $\rightarrow$ FMCW, 1 $\rightarrow$ CW • Bit 1: TD data in [V] • Bit 2: Magnitude of frequency-domain (FD) data in [dB] • Bit 3: Phase of FD data in [deg] • Bit 4: Object angle in [deg] • Bit 5: Human Tracker data • Bit 6: Range Finder data
Bin Size [mm]	float32	range resolution
Number of Samples	uint16	number of range bins
Threshold Filter Factor	uint16	see section 3.8
Over Threshold Factor	uint16	see section 3.8
Minimum Target Separation [Bins]	uint16	see section 3.8
Minimum Target Distance [m]	uint16	see section 3.8
Maximum Detection Losses	uint16	see section 3.8
Maximum Range Shift [m]	uint16	see section 3.8
Enable Tracker	uint16	see section 3.8
Maximum Number of Targets	uint16	see section 3.8
Background Filter Factor	uint16	see section 3.8

Table 3.8: Header of Human Tracker binary files

Parameter	Data Type	Description
File creation Time [ms]	uint64	Epoch Unix time format
Radar Number	uint32	unique identifier of the radar module
Start-Frequency [MHz]	uint16	FMCW: start frequency, CW: Tx frequency
Stop-Frequency [MHz]	uint16	FMCW stop frequency, CW: not used
Ramp Time [ms]	uint16	FMCW: ramp duration, CW: sampling time
Attenuation [dB]	float32	Tx attenuation during measurement
Data Format	uint32	Binary mask of the stored measurement data: • Bit 0 (LSB): 0 $\rightarrow$ FMCW, 1 $\rightarrow$ CW • Bit 1: TD data in [V] • Bit 2: Magnitude of frequency-domain (FD) data in [dB] • Bit 3: Phase of FD data in [deg] • Bit 4: Object angle in [deg] • Bit 5: Human Tracker data • Bit 6: Range Finder data
Bin Size [mm]	float32	range resolution
Number of Samples	uint16	number of range bins
Measurement Type	uint16	1: Up-Ramp, 2: Down-Ramp
Maximum Acceleration	uint16	see section 3.9
Minimum Distance [m]	uint16	see section 3.9
Detection Threshold	int16	see section 3.9
Maximum Uncertainty	uint16	see section 3.9
Tracking Enabled	uint16	0: disabled, 1: enabled

Table 3.9: Header of Range Finder binary files

Binary Files saved from Human Tracker

The amount of data measured by the Human Tracker is highly dependent on the parameters set and the environment, so it may vary in each measurement. Therefore each measurement is headed by the time in milliseconds given as 64-bit unsigned integer value and the number of targets found given as 16-bit unsigned integer value which indicates the amount of data in this measurement. If no targets were found, the value for the number of targets is zero. For each target the following values are stored in a measurement:

- *Target ID (uint16)*
- *Target tracked (uint16)*
- *Detection counter (uint16)*
- *Level [dBm] (float32)*
- *Distance [m] (float32)*
- *Angle [deg] (float32)*

Example

In the first step, the header data must be extracted from file (see table 3.8). Now, we can read the rest of the binary data stream:

while not end-of-file:

```
    read measurement time-stamp in [ms]
    read number of found targets
    for (n = 1 to number of targets)
        read Target ID
        read Target tracked
        read Detection counter
        read Level [dBm]
        read Distance [m]
        read Angle [deg]
```

Binary Files saved from Range Finder

The data stream of a Range Finder binary file that follows the header shown in table 3.9 is quite simple. For each measurement the following values are stored:

- *Radar Time [ms] (uint64)*
- *Range [m] (float32)*
- *Angle [deg] (float32)*
- *Velocity [m/s] (float32)*

The Radar Time corresponds to the Epoch Unix time in milliseconds and is measured inside the radar module. On power up this time starts with zero but it can be set to the current time value. Therefore see the Interface Description /CMD/.

Example

In the first step, the header data must be extracted from file (see table 3.9). Now, we can read the rest of the binary data stream:

```
while not end-of-file:
    read radar time-stamp in [ms]
    read range [m]
    read angle [deg]
    read velocity [m/s]
```

3.10.6 Data Replay Mode

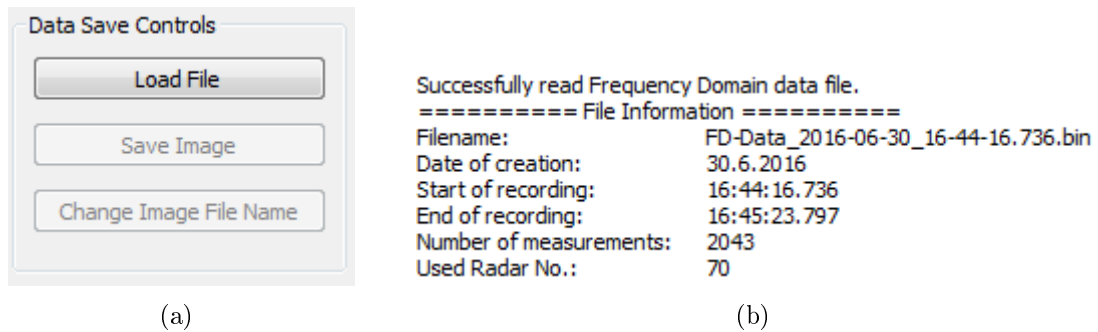


Figure 3.19: (a) Data Save Controls while disconnected, (b) Data file information

In the data replay mode previous saved files can be played again and analyzed. Please note that this feature works **only for binary files**. To start this mode, a binary file has to be loaded by using the button *Load File*. This button is only available when the IMST SenTool is in the disconnected state. Figure 3.19a shows the *Data Save Controls* in this state. When the *Load File* button is pressed, a dialog appears where a binary file can be searched and selected.

When a binary file is selected, it is checked for errors and loaded into memory. If the reading was successful, a message appears in the *Message Window* containing some information about the file. Figure 3.19b shows an example with information about a read frequency domain data file.

The successful reading of a binary data file also modifies some controls of the GUI. The tab corresponding to the data type of the loaded file (*Data Plot*, *Human Tracker* or *Range Finder*) is selected and the plot of the first measurement is shown in the tab specific plot. Additionally, some controls of the *Radar Controls* are highlighted, showing the parameters found in the file header. Depending on the data type in the loaded file, some controls become available in the *Data Plot Controls*, *Human Tracker Controls* or *Range Finder Controls*. Also the *Polar Plot* and *History Plot* could become available if the data type is

compatible. The *Save Image* and *Change Image File Name* buttons also become available to save screenshots.



Figure 3.20: Status bar after successful reading a file

The most significant difference is the change in the *Status Bar* where some new controls appear after a data file has been loaded (see figure 3.20). The enumerated points are described below:

1. Here some short information appears when the mouse moves over a available control.
2. Control buttons:
 -   Play/Pause button to start/pause playing the data. The replaying speed corresponds to the original measurement speed.
 -  Stop button to stop playing and reset the progress.
 -   Previous/Next buttons to switch to the previous/next measurement.
3. Slider that shows the progress. When grabbed with the mouse it can be scrolled through the measurements.
4. Progress information that shows the actual time of the measurement in relation to the total time. The values in brackets show the current measurement number relative to the total number of measurements.

3.11 Troubleshooting

The following actions may be useful in case of errors or unexpected radar behavior:

- Reconnect the radar module,
- Switch the radar module OFF and ON,
- Restart the IMST SenTool,
- Delete the configuration file `Config.cfg` in the GUI source directory and restart the radar module as well as the IMST SenTool.

4. Technical Specifications

4.1 Datasheet

Parameter	Specification
<i>General</i>	
Modulation	CW / FMCW
Operating Frequency	24.00 GHz – 24.25 GHz (@ ISM band)
Number of Channels	1 Transmit-Channel (Tx), 2 Receive-Channels (Rx) with I/Q demodulator for each channel
Data Interface	SPI, CAN, Ethernet (with PoE)
Certification	CE Approval (Ethernet version only)
<i>Antenna</i>	
Antenna Type	Integrated Patch Antenna
Antenna Characteristics	65° Azimuth and 24° Elevation
Antenna Polarization	linear
<i>Measurement</i>	
Measurement Range	0.6 m – 307 m (@ ISM band)
Range Resolution	max. 0.6 m (@ ISM band)
Measurement Speed	±3200 m/s (theoretical value @ 24 GHz)
Speed Resolution (1 – 100 ms)	6.25 m/s – 6.25 cm/s (@ 24 GHz)
Angle Measurement Accuracy	2° – 3°
<i>Operation Parameters</i>	
Frequency Ramp Duration	1 ms – 100 ms
Update Rate	typ. 10 Hz – 200 Hz (see section 4.2)
EIRP Output Power	typ. 10 dBm – 19 dBm (tunable)
Operating Temperature	min. -40 °C, max. 60 °C (see section 4.3)
<i>Power Supply</i>	
Operation Voltage	10.5 V – 13 V, 44 V – 54 V PoE
Standby Power	3.0 W
Operating Power	4.5 W

Parameter	Specification
<i>Housing</i>	
Dimension (L × W × H)	98.0 mm × 87.0 mm × 42.5 mm (Housing) 114.0 mm × 87.0 mm × 42.5 mm (with Bushing)
Weight	280 g
Mounting	4 Mounting Holes (Ø 5 mm)
Cable and Connector	Shielded 9 Wire Cable (SPI/CAN) CAT-6 with RJ45 Female Plug (Ethernet)
Protection Code	IP 65

4.2 Update Rate

The maximum possible measurement update rate depends on many parameters as ramp duration, signal processing time and the amount of transmitted data samples. Therefore, this value is exemplarily given for three typical applications. In the following examples, the use of the ISM band and a maximum distance of 30 m is assumed:

- FMCW measurement and transmission of TD data from 4 channels → 15.5 Hz,
- FMCW measurement and transmission of FD data (amplitude and object angle) from 2 channels → 98 Hz,
- FMCW measurement and transmission of FD data (amplitude only) from 1 channel → 180 Hz.

4.3 Operating Temperature

The maximum possible operating temperature mainly depends on the duty cycle of the RF transmitter (please note that the duty cycle is always 100% if the *Power Save* mode is disabled). The following table shows the internal temperature of the radar module at an environment temperature of 60 °C for different duty cycles:

Duty Cycle	Radar Temperature
1%	81 °C
10%	82 °C
25%	83 °C
50%	84 °C
100%	86 °C (→ overheat)

5. Radar Fundamentals

The following chapter provides a short overview about the basics of radar technology employed in the IMST's sentireTM Radar Module. For more details and applications the reader is asked to refer to technical literature.

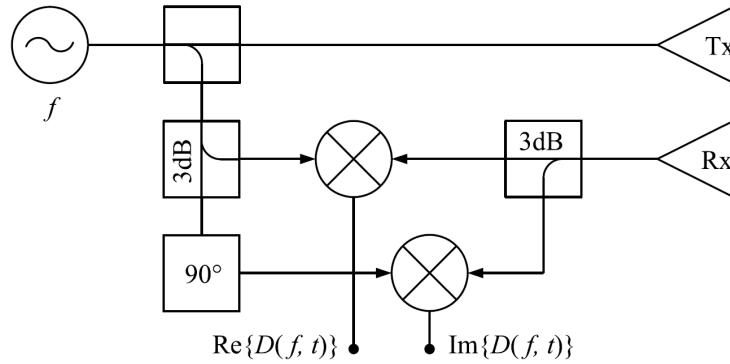


Figure 5.1: Generalized block diagram of a bistatic CW/FMCW radar system with one transmitter and one receiver channels and I/Q demodulator

5.1 Principle of Operation

The operation principle of a typical CW/FMCW radar system is demonstrated in figure 5.1 and will be described below. A radar consists of the following components:

- Signal source for generation of a (frequency-modulated) signal,
- Transmitter antenna (Tx),
- One or several receiver antennas (Rx),
- One or several mixers for generation of the Intermediate Frequency (IF),
- Digital sampling unit (A/D converter),
- Couplers, filters, amplifiers, phase shifters (these components are ignored here).

Assuming that the Tx antenna transmits a signal with the frequency f_T and the Rx antenna at the same time receives the frequency f_R , the resulting intermediate frequency f_I can be calculated using the formula:

$$f_I = f_T - f_R. \quad (5.1)$$

It is obvious that the value of f_I can be positive as well as negative (depending on f_T and f_R), which explains the need for the I/Q demodulator in figure 5.1. Nevertheless, the following calculation will be based on real-valued sampling. Now we assume that the IF signal is digitized with a sampling frequency f_S for the time interval T . This yields a discrete Time-Domain (TD) signal with

$$N_S = T \cdot f_S \quad (5.2)$$

real data samples, which can be transformed to Frequency-Domain (FD) using a Fast Fourier Transform (FFT). The result is a complex-valued signal consisting of $N_S/2$ data samples with a frequency step of

$$\delta_f = \frac{1}{T}. \quad (5.3)$$

This expression represents the system's frequency resolution and allows to assign a data sample on position n (also called "bin number n " with $n \geq 0$) of the IF signal spectrum to a certain intermediate frequency:

$$f_I(n) = \delta_f \cdot n. \quad (5.4)$$

The maximum intermediate frequency f_{max} which can unambiguously be measured by the system, is calculated using the equations 5.2 to 5.4:

$$f_{max} = \delta_f \cdot \frac{N_S}{2} = \frac{N_S}{2 \cdot T} = \frac{T \cdot f_S}{2 \cdot T} = \frac{f_S}{2}. \quad (5.5)$$

Equation 5.5 simply states that the sampling frequency f_S must be twice as large as highest frequency f_{max} included in the IF signal. This is nothing else than the representation of the Nyquist-Shannon sampling theorem. In the IMST's sentireTM Radar Module, the TD signal always consists of 1024 data samples. Hence, the corresponding maximum frequency is given by the equation:

$$f_{max} = \frac{512}{T}. \quad (5.6)$$

5.2 FMCW Mode

If operating in FMCW mode, the radar generates a linear frequency ramp of the duration T covering the bandwidth B (see figure 5.2). Assuming a point scatterer at a distance R , the propagation time from transmitter to the receiver can be expressed as:

$$T_0 = \frac{2 \cdot R}{c_0} \quad (5.7)$$

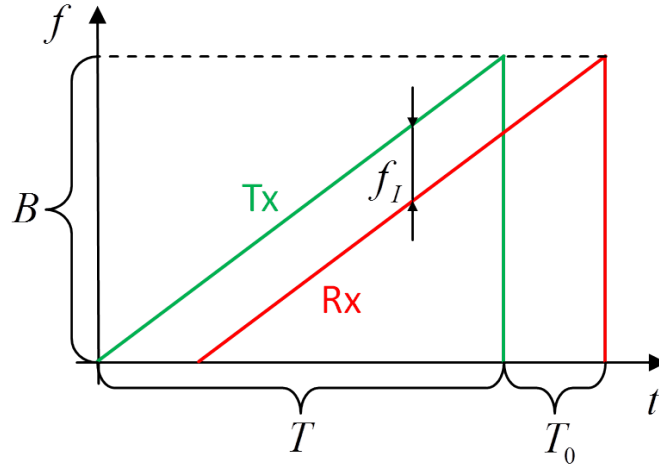


Figure 5.2: Operation principle of an FMCW radar

with c_0 representing the speed of light. Thus, the signal arriving at the receiver antenna (Rx) is delayed by the time T_0 compared with the signal at the transmitter antenna (Tx). Consequently, the frequencies f_T and f_R appearing at the mixer inputs are also slightly different. This difference exactly corresponds to the intermediate frequency f_I (compare to equation 5.1) and can easily be calculated using figure 5.2:

$$f_I = \frac{B \cdot T_0}{T} = \frac{2 \cdot B \cdot R}{T \cdot c_0}. \quad (5.8)$$

This gives us the possibility to assign any target distance R to a fixed intermediate frequency f_I :

$$R(f_I) = \frac{T \cdot c_0}{2 \cdot B} \cdot f_I. \quad (5.9)$$

The corresponding range resolution δ_r of the radar can be derived from equation 5.9 using the frequency resolution δ_f known from equation 5.3:

$$\delta_r = R(\delta_f) = \frac{T \cdot c_0}{2 \cdot B} \cdot \delta_f = \frac{c_0}{2 \cdot B}, \quad (5.10)$$

which is the well-known formula valid for *all* radar sensors. Analyzing FD data from an FMCW measurement, it is also possible to say that the “range bin number n ” corresponds to a target distance of:

$$R(n) = \delta_r \cdot n. \quad (5.11)$$

The maximum range which can (theoretically) be measured by the radar, is calculated analogue to equation 5.10 using the maximum frequency f_{max} from equation 5.5:

$$R_{max} = R(f_{max}) = \frac{T \cdot c_0}{2 \cdot B} \cdot f_{max} = \frac{c_0 \cdot N_S}{4 \cdot B}. \quad (5.12)$$

For the special case of the IMST Radar Module ($N_S = 1024$), the maximum measurement range corresponds to

$$R_{max} = \frac{c_0 \cdot 256}{B}. \quad (5.13)$$

5.3 Doppler Mode

If operating in CW (or Doppler) mode, the radar transmits a single frequency f_0 without any modulation, i.e. with a bandwidth of $B = 0$. From equation 5.10 it becomes clear, that no range resolution is possible in this case, as the transmitted and received frequencies (f_T and f_R) are equal for all *static* targets. Anyway, this behavior significantly changes for *moving* objects: each radial velocity v between radar and target causes a frequency shift of the reflected signal, the so called Doppler frequency

$$f_D = \frac{2 \cdot f_0 \cdot v}{c_0}. \quad (5.14)$$

For the sake of completeness, it should be mentioned that equation 5.14 is only valid for non-relativistic cases ($|v| \ll c_0$), which is usually fulfilled for the most realistic applications. For a target moving *away* from sensor (v is defined as positive), we expect to receive a signal with the frequency

$$f_R = f_T - f_D, \quad (5.15)$$

leading to an intermediate frequency of:

$$f_I = f_T - f_R = f_D = \frac{2 \cdot f_0 \cdot v}{c_0}. \quad (5.16)$$

Again, we can establish a link between a certain target velocity v and the corresponding intermediate frequency f_I :

$$v(f_I) = \frac{c_0}{2 \cdot f_0} \cdot f_I. \quad (5.17)$$

Analogue to equations 5.10 and 5.12, the velocity resolution δ_v as well as the maximum velocity v_{max} can be calculated by inserting δ_f and f_{max} into equation 5.17:

$$\delta_v = v(\delta_f) = \frac{c_0}{2 \cdot f_0} \cdot \delta_f = \frac{c_0}{2 \cdot f_0 \cdot T}. \quad (5.18)$$

$$v_{max} = v(f_{max}) = \frac{c_0}{2 \cdot f_0} \cdot f_{max} = \frac{c_0 \cdot N_S}{4 \cdot f_0 \cdot T}. \quad (5.19)$$

For the special case of the IMST Radar Module ($N_S = 1024$), the maximum unambiguously measured target velocity corresponds to

$$v_{max} = \frac{c_0 \cdot 256}{f_0 \cdot T}. \quad (5.20)$$

In comparison to FMCW measurements, an additional difficulty appear in the CW (or Doppler) mode. Due to equation 5.18, it is possible to say that the “velocity bin number n ” corresponds to a radial target velocity of

$$v(n) = \delta_v \cdot n. \quad (5.21)$$

Unfortunately, this statement doesn't take into account the direction of the target motion: positive as well as negative velocities of the same magnitude appear at an identical position in the IF signal spectrum. To resolve this ambiguity, the I/Q demodulator of the radar sensor (see figure 5.1) can be used. Positive and negative Doppler frequencies can easily be separated by using complex instead of real-valued IF signals. The mathematical background of this operation can be extracted from Fourier Transforms theory.

5.4 Range-Doppler-Ambiguity

In sections 5.2 and 5.3, the target distance R as well as its radial velocity v have been linked to the intermediate frequency f_I measured by the radar sensor (compare equations 5.8 and 5.16). If dealing with moving targets in FMCW operation mode, this leads to an effect called Range-Doppler-Ambiguity. The main problem is the “mixing” of two independent parameters (R and v) to a single measurement value (f_I), resulting in erroneous range measurements.

The expected range error appearing at a target moving with a radial velocity v can be estimated by superposition of equations 5.8 and 5.16:

$$\Delta R(v) = \frac{f_0 \cdot T}{B} \cdot v. \quad (5.22)$$

In case of spatially well separated targets, this limitation can be overcome by alternating transmission of rising (Up) and falling (Down) frequency ramps. If the same target can be localized in both data sets, its parameters R and v can unambiguously be reconstructed by measuring the corresponding position in the Up and Down signal spectra. In case of noisy data with many spatially close targets, a unique reconstruction of target parameters is usually not possible.

5.5 Angle Measurement

The IMST's sentireTM Radar Module is equipped with two independent receiver antennas for measuring the Angle-of-Arrival of a received signal. This feature can be used to generate a 2-dimensional reflectivity map of the illuminated scene. To derive the formula for the view angle extraction, two receiver antennas separated by a baseline b are assumed (see figure 5.3). Further, a plane phase front shall arrive from the direction defined by the angle α . Consequently, the incident wave arrives at antenna Rx1 a little bit later compared with antenna Rx2. With the center frequency f_c , this causes a phase shift φ between both receiver antennas, given by the formula:

$$\varphi = \frac{2\pi \cdot f_c \cdot d}{c_0}. \quad (5.23)$$

The parameter d describes the additional distance the wave must travel to reach the antenna Rx1. This value is defined by

$$d = b \cdot \sin(\alpha), \quad (5.24)$$

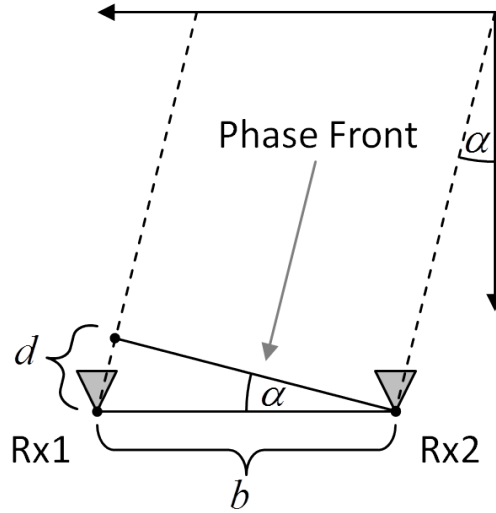


Figure 5.3: Estimation of the View Angle using two receiver antennas

so we can expect a phase difference of

$$\varphi = \frac{2\pi \cdot f_c \cdot b \cdot \sin(\alpha)}{c_0} \quad (5.25)$$

between the signals received by the antennas Rx1 and Rx2. Equation 5.25 can be inverted to derive the final relation between the measured phase difference φ and the angle α :

$$\alpha(\varphi) = \arcsin\left(\frac{\varphi \cdot c_0}{2\pi \cdot f_c \cdot b}\right). \quad (5.26)$$

Finally, some physical limitations must be added to equation 5.26:

- the assumption of a plane phase front (see figure 5.3) is only valid for targets located in the far-field of the antenna,
- to avoid ambiguities, the argument of the $\arcsin()$ function must be in limits $[-1, 1]$,
- as the baseline b is defined for lumped antenna elements, the formula gets imprecise for large angles α ,
- non-ideal hardware components, as transmission lines, sensor housing, radome or PCB ground planes, additionally influence the antenna behavior as well as the estimated view angles.

In the special case of the IMST Radar Module, a baseline of 6.25 mm has been selected, corresponding to a half wave length at 24 GHz.

Appendix

A Referenced Documents

The following table contains a list of documents referenced by this manual.

Ref. ID	Document Name	Source
/RAD/	sR1200_Radar_Manual.pdf	IMST GmbH
/CMD/	sR1200_Command_Interface.pdf	IMST GmbH
/GUI/	sR1200_Development_Kit.pdf	IMST GmbH

B List of Abbreviations

ASCII:	American Standard Code for Information Interchange
CAN:	Controller Area Network
CE:	European Community
CW:	Continuous Wave
DC:	Direct Current
DSC:	Digital Signal Controller
EEPROM:	Electrically Erasable Programmable Read-Only Memory
EIRP:	Equivalent Isotropically Radiated Power
ELV:	Extra-Low Voltage
ESD:	Electrostatic Discharge
FD:	Frequency Domain
FFT:	Fast Fourier Transform
FMCW:	Frequency Modulated Continuous Wave

FTDI:	Future Technology Devices International Ltd. (company name)
GUI:	Graphical User Interface
HDD:	Hard Disk Drive
HT:	Human Tracker (algorithm)
ID:	Identifier
IEC:	International Electrotechnical Commission
IEEE:	Institute of Electrical and Electronics Engineers
IF:	Intermediate Frequency
IP:	Internet Protocol
ISM:	Industrial, Scientific and Medical (frequency band)
JPEG:	Joint Photographic Experts Group (file format)
LAN:	Local Area Network
LED:	Light-Emitting Diode
MAC:	Media Access Control
MB:	Megabyte (corresponds to 1.000.000 bytes)
PoE:	Power over Ethernet
RCS:	Radar-Cross-Section
RF:	Radio Frequency or Range Finder (algorithm)
Rx:	Receiver (antenna, etc.)
SPI:	Serial Peripheral Interface
TD:	Time Domain
Tx:	Transmitter (antenna, etc.)
USB:	Universal Serial Bus

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Important Notices

Regulatory Compliance Information

The use of radio frequencies is limited by national regulations. The Radar Module has been designed to comply with the European Union's R&TTE (Radio & Telecommunications Terminal Equipment) directive 1999/5/EC and can be used license free within the European Union at 24–24.25 GHz. Nevertheless, please refer to your national frequency regulation if there is any need for a notification at least.

The Radar Module (with housing) has been designed to be used with suitable IT-equipment to collect, store and process the radar data (referred as “final system”). According to the R&TTE directive, the declaration of compliance with the essential requirements of the R&TTE directive is within the responsibility of the manufacturer of the final system. A declaration of conformity (DoC) for the Radar Module is available from IMST GmbH on request.

The applicable regulation requirements are subject to change. IMST GmbH does not take any responsibility for the correctness and accuracy of the aforementioned information. National laws and regulations, as well as their interpretation can vary with the country. In case of uncertainty, it is recommended to contact either IMST's accredited Test Center or to consult the local regulatory authority of the respective country of use.

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Contact Information

IMST GmbH
Carl-Friedrich-Gauss-Str. 2-4
47475 Kamp-Lintfort
Germany
Phone: +49 2842 981 0
Fax: +49 2842 981 199
E-Mail: radar@imst.de
Internet: www.radar-sensor.com